

# MTH202 900+ MIDTERM MCQS

## SEARCHING FILE FOR GRAND QUIZ

## CREATE BY ORANGE MONKEY TEAM

1<sup>st</sup> file 95% confirm

1. A function from B to A is called onto function if its range is:

A

2. If  $p = \text{It is raining}$   $q = \text{She will go to college}$

$q \wedge p$

3. A subset  $f$  of  $B \times A$  is said to be a function from B to A if the domain of  $f$  is B and the first element of order pairs of  $f$ :

Does not repeat

4. The negation of "Today is Monday" is:

Today is not Monday

5. A function in which the second element of order pair is distinct is called:

One-one function

6. The converse of the conditional statement  $q \rightarrow p$  is:

$p \rightarrow q$

7. Letters of SAP taken all at a time can be written in:

6 ways

8. Contra-positive of the given statement "If two angles are congruent, then they have the

same measure” is:

If two angles do not have the same measure, then they are not congruent

9. A student has a maximum of 720 words from a combination of letters of a word given word is:

CANADA

10. The statement “It is not raining if and only if roads are dry” is logically equivalent to:  
Roads are dry if and only if it is not raining

11. The inverse of this statement If p, then q is:  
If not p, then not q

12. A converse of this statement If p, then q is:  
If q, then p

13. How many bytes are required to encode 2000 bits of data:  
2

14. A collection of graph is:  
vertices and columns

15. The number of edges in a complete graph with ‘n’ vertices is equal to:  
 $n(n-1)/2$

16. Error correcting code is a  
hamming code

17. The symbol II is ASCII stands for:  
American Standard Code for Information Interchange

18. What is domain of function  $f(x) = x^{1/2}$ :  
 $[0, \infty)$

19. ordered collection of objects is:  
set

20. A function is a Domain of:  
the maximal set of numbers for which a function is defined

21. Range of a function is :  
the maximal set of numbers which a function can take values

22. In an undirected graph the number of nodes with odd degree must be:  
Even

23. What is the cardinality of the set of odd positive integers less than 10?  
5

24. The Gray code of a number whose binary representation is 1000 is:

1100

25. The domain of a relation R from A to B is symbolically written as:

$\text{Dom}(R) = \{a \in A \mid (a, b) \in R\}$

26. The number of element is the power set of P(not empty set) denoted by P(not empty set)) is

Not empty set

27. The number of element is the power set of P(not empty set) denoted by P(P(P(not empty set))) is

4

28. Let A be a non-empty set and P(A) the power set of A. Define the 'subset' relation,  $\underline{C}$ , as follows; for all X,Y belong to P(a),  $\underline{C}Y \leftrightarrow \text{FOR ALL } x, \text{ iff } x \text{ belong to } X, \text{ Then } \underline{C}$  is

$\underline{C}$  is a partial order relation

29. X belongs to A and x belongs to B, there x belongs to

$A \cup B$

30.  $A \cap B =$   $\{x \in U \mid x \in A \text{ and } x \in B\}$

31.  $A \cup B =$

$\{x \in U \mid x \in A \text{ or } x \in B\}$

32. If  $A \cap B =$  not empty set then A and B are called

Overlapping

33. Let t be tautology, then  $\sim(\sim t)$  is

t

34. Let the set  $A = \{1, 2, 3, 4\}$  Then the relation  $\{(2, 4), (4, 2)\}$  is

Symmetric

35. Sets are denoted by

Capital letter

36. The objects are

Elements or members of the set

37. The elements of a set are represented by \_\_\_\_\_.

Lower case letter

38. Let  $p$  be true and  $q$  be false, then  $p \wedge \sim q$  is \_\_\_\_\_.

True

39. If  $f(x) = x$  and  $g(x) = -x$  are both one to one functions then  $(f + g)(x)$  is also one to one function.

True

40. In the digital circuits, ON represents \_\_\_\_\_.

True or 1

41. Let  $R$  and  $S$  be reflexive relations on a set  $A$  then  $R \cap S$  is reflexive.

True

42. There is at least one loop in the graph of an irreflexive relation

False

43. There is no loop in the graph of an irreflexive relation.

True

44. Complement of  $U$  is \_\_\_\_\_.

$\emptyset$

45. Let  $p =$  I am hard working,  $q =$  I am intelligent, Then  $\sim p \wedge q$  means \_\_\_\_\_

I am not hardworking But I am intelligent

46. IF  $(p \text{ XOR } q)$  is True, then \_\_\_\_\_.

$p$  is true,  $q$  is true

47. Which of the following is a right biconditional statement.

It is hot outside and I will buy an ice cream...

48. Truth value of  $(3+1=4) \wedge (5 \text{ is an odd number})$  is \_\_\_\_\_.

False

49. In the matrix representation of an irreflexive relation all the entries in the main diagonal are.....

0

50. Vertical line test is used to determine that whether the graph of a relation is a function or not.

True

51. IF  $p \rightarrow q$  is false then  $\sim p \rightarrow \sim p$  will be \_\_\_\_\_.

False

52. A union  $B = B \cup A$ ,  $A \cap B = B \cap A$ , are called

**Commutative laws**

53. Let  $R$  be a binary relation on a set  $A$ .  $R$  is anti-symmetric iff \_\_\_\_\_.

$a, b \in A$  if  $(a, b) \in R$  and  $(b, a) \in R$  then  $a = b$ .

54. Let  $R$  be a relation on a set  $A$ .  $R$  is symmetric if, and only if,

$a, b \in A$ , if  $(a, b) \in R$ , then  $(b, a) \in R$ .

55.  $R$  is not reflexive iff there is an element " $a$ " in  $A$  such that  
 $(a, a) \notin R$

56. In the matrix representation of an reflexive relation all the entries in the main diagonal are.....

1

57.  $R$  is not transitive iff there are elements  $a, b, c$  in  $A$  such that

if  $(a, b) \in R$  and  $(b, c) \in R$  but  $(a, c) \notin R$ .

58. For the null relation  $\emptyset$  on  $A$  to be symmetric, it must satisfy the implication

if  $(a, b) \in \emptyset$  then  $(b, a) \in \emptyset$  is true by default, since the condition  $(a, b) \in \emptyset$  is always false

59. The universal relation  $A \times A$  is symmetric, for it contains all ordered pairs of elements of  $A$ . Thus,

if  $(a, b) \in A \times A$  then  $(b, a) \in A \times A$  for all  $a, b$  in  $A$ ....symmetric

60. The null relation  $\emptyset$  on  $A$  is transitive, because the implication.

if  $(a, b) \in \emptyset$  and  $(b, c) \in \emptyset$  then  $(a, c) \in \emptyset$  is true by default, since the condition  $(a, b) \in \emptyset$  is always false

61. The universal relation  $A \times A$  is transitive for it contains all ordered pairs of elements of  $A$ .

if  $(a, b) \in A \times A$  and  $(b, c) \in A \times A$  then  $(a, c) \in A \times A$

62. Let  $A$  be subset of universal set  $U$  then  $A \cup A^c =$

$U$

63. Let  $A$  be subset of universal set  $U$  then  $A \cap A^c =$

$\emptyset$

64. Let  $A$  be subset of universal set  $U$  then  $A^c =$

65.  $\sim p$  is True when  $\underline{p \text{ is False}}$ .

P is false

66. Let R and S be reflexive relations on a set A then  $R \cap S$  is reflexive.

True

67. If  $p = T, q = T, r = T$ , then  $r \leftrightarrow$  will be \_\_\_\_\_.

T

68. Which of the logical expressions represents the sentence you read stories daily, then your vocabulary will increase and conversely. ?

$p \rightarrow q$

69. P is sufficient for q means ' $p \rightarrow q$ '

True

70. "p only if q"  $p \rightarrow q$ .

True

71. if p, q"  $p \rightarrow q$  is

True

72. "p implies q"  $p \rightarrow q$

True

73. if p then q"  $p \rightarrow q$

True

74. "not p unless q"  $p \rightarrow q$

True

75. q follows from p"  $p \rightarrow q$

True

76. "q if p"  $p \rightarrow q$

True

77. "q whenever p"  $p \rightarrow q$

True

78. "q is necessary for p"  $p \rightarrow q$

True

79. The function defined from Z to Z as  $f(x) = 1/(x+2)(x-2)$  is not well defined

because

Function is not defined at  $x = -2$  and  $x = 2$

80. The function defined from  $\mathbb{Z}$  to  $\mathbb{Z}$  as  $f(n) = \pm n$  is not well defined because  
f is not well defined since each integer  $n$  has two images  $+n$  and  $-n$

81. The function defined from  $\mathbb{Z}$  to  $\mathbb{Z}$  as  $f(n) = \text{square root } n$  is not well defined because  
f is not defined for  $n < 0$  since f then results in imaginary values (not real)

82. The function defined from  $\mathbb{Z}$  to  $\mathbb{Z}$  as  $f(n) = \text{square root}(n^2+1)$  is not well defined because  
f is well defined because each integer has unique (one and only one) image in  $\mathbb{R}$   
under f.

83. Generally, ( s AND t ) is \_\_\_\_\_, where t is tautology

S

84. Which of the following is not a binary operation on the set of integers?

- Or subtraction.....both

85. Which of the following are binary operations on the set of natural numbers  $\mathbb{N}$ .

+ and . or addition or multiplication.....both

86. Which of the following is a binary operation on  $\mathbb{Z}$ , the set of integers.

- Or subtraction ...both

87. Which of the following is a binary operation on the set of non-zero rational numbers  $\mathbb{Q} - \{0\}$  .

Division

88. Which of the following is not a binary operation on  $\mathbb{Z}$ .

Division

89. If the conclusion is True for the true premises, then the argument is called as \_\_\_\_\_.

Valid

90. If the conclusion is false for the true premises, then the argument is called as \_\_\_\_\_.

Invalid

91.  $A = \{1, 2\}$   $B = \{3, 4\}$ ,  $R = \{(1, 3) (2, 4)\}$ . Then the complement of R is \_\_\_\_\_.

$\{(1, 4) (2, 3)\}$

92. Let  $p = 1$  am not a student, then  $\sim(\sim p)$  represents \_\_\_\_\_.

I am not a student

93. Let  $p \rightarrow q$  be a conditional statement, then the statement  $q \rightarrow p$  is called.

Converse

94. The switches in series acts just like \_\_\_\_\_.

AND gate

95.  $Y=x^3$  is a graph of bijective function from  $R$  to  $R$

True

96. Let  $A, B, C$  be subset of universal set  $U$  then the distributive Law is ...

$$A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$$

$$A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$$

97. Composite relation symbolically written as \_\_\_\_\_.

$$\text{SoR} = \{(a,c) | a \in A, c \in C, \exists b \in B, (a,b) \in R \text{ and } (b,c) \in S\}$$

98. Inverse of a function may not be a function.

True

99. Let  $f$  and  $g$  be two function defined by  $f(x) = x+2$  and  $g(x) = 2x+1$ . THEN the composition of  $f$  and  $g$  is \_\_\_\_\_.

$$2x^2 + 5x + 2$$

100. Truth value of  $p \vee q$  will be

101. The set of natural number is equal to \_\_\_\_\_.

The set of positive integers

102. Let  $R$  be a binary relation on a set  $A$ . If  $R$  is anti symmetric then \_\_\_\_\_.

Inverse of  $R$  is antisymmetric

103. Let  $R$  be a binary relation on a set If  $R$  is transitive, then

Inverse of  $R$  is transitive

104. Let  $R$  be a binary relation on a set If  $R$  is symmetric, then

Inverse of  $R$  is symmetric

105. Let  $R$  be a binary relation on a set If  $R$  is reflexive, then

Inverse of  $R$  is reflexive

106. What is the truth value of the sentence ? It rains if and only if there are clouds.



**False**

**107. What is the truth value of the sentence ? It rains if and only if it is a weekend day.**

**True**

**108. What is the truth value if the sentence?. You buy an ice cream cone if and only if it is hot outside**

**True**

**109. What is the truth value of the sentence? You win the contest if and only if you hold the only winning ticket.**

**True**

**110. What is the truth value of the sentence?. You will be informed if and only if you read the news paper every day**

**True**

**111. What is the truth value of the sentence?. The train runs late if and only if it is a day I take the train.**

**True**

**112. What is the truth value of the sentence?. This number is divisible by 6 if and only if it is divisible by both 2 and 3.**

**True**

**113. If  $p \rightarrow q$  is false then Truth value of  $p \leftrightarrow q$  is**

**True**

**114. The logic Gate OR and AND are uniary operation  $\{0,1\}$**

**False**

**115. The logic gate NOT is a uniary operation on  $\{0,1\}$**

**True**

**116. The logic gates OR and AND are binary operations on  $\{0,1\}$**

**True**

**117. The logical connectives  $\vee, \wedge, \rightarrow, \leftrightarrow$  are binary operations on the set  $\{T, F\}$**

**True**

118. The set operations union  $\cup$ , intersection  $\cap$  and set difference  $-$ , are binary operators on the power set  $P(A)$  of any set  $A$ .

True

119. The logical expression  $p \leftrightarrow q$  is equal to \_\_\_\_\_.

$(p \rightarrow q) \wedge (q \rightarrow p)$

120. A function whose inverse function exists is called /an

invertible function.

121. Let  $X = \{1, 5, 9\}$  and  $Y = \{3, 4, 7\}$ . Define  $g: X \rightarrow Y$  by specifying that  $f(1) = 7$ ,  $f(5) = 3$ ,  $f(9) = 4$  then which of the following statement about  $f$  is true

$f$  is both one-to-one and onto

122. Which of the following is not true for an identity function?

The identity function is always a constant function

123. Let  $p =$  I am hard working, the  $\sim(\sim p)$  represents \_\_\_\_\_.

I am hard working

124. Which of the following is not a representation of a relation?

Venn diagram

125. Common ratio in sequence '36, 12, 4, 4/3, ....' is .....

1/3

126. Common ratio in sequence 1, 2, 4, 8, 16, ...

2

127. Common ratio in sequence 3, - 3/2, 3/4, - 3/8, ...

-1/2

128. Common ratio in sequence 0.1, 0.01, 0.001, 0.0001, ...

0.1 or 1/10

129. if  $x \equiv 17 \pmod{5}$  Which of the following integers are valid solution for  $x$ ?

12

130. A set is called finite if, and only if, it is the \_\_\_\_\_ or there is \_\_\_\_\_

Empty set or onto

131. If a relation  $R=(1,2)(2,3)(3,4)(4,1)(2,2)$  is given then which of the following is true about

$R$  is anti symmetric

132. The statement ‘ Islamabad is the capital of Pakistan AND abu dhabi is the capital of U.A.E is a \_\_\_\_\_.

Conjunction

133. Let  $A$  and  $B$  be subset of universal set  $U$  then  $A-(A-B) =$

$$A \cap B$$

134.  $P \leftrightarrow q$  is true when \_\_\_\_\_.

$P$  is false,  $q$  is false

135. Let  $A = \{1,2,3,...,50\}$  and  $B=\{2,4,6,8,10\}$ . Then the Cartesian product of  $A$  and  $B$  has \_\_\_\_\_ elements.

250

136. Cartesian product of  $A$  and  $B$ , denoted by  $A \times B$  (read as “A cross B”) is the set of all ordered pairs  $(a, b)$ , where  $a$  is in  $A$  and  $b$  is in  $B$ .

$$A \times B = \{(a, b) | a \in A \text{ and } b \in B\}$$

137. The domain of a relation  $R$  from  $A$  to  $B$  is the set of all first elements of the ordered pairs which belong to  $R$  denoted by  $\text{Dom}(R)$ .

$$\text{Dom}(R) = \{a \in A | (a, b) \in R\}$$

138. The range of a relation  $R$  from  $A$  to  $B$  is the set of all second elements of the ordered pairs which belong to  $R$  denoted  $\text{Ran}(R)$ .

$$\text{Ran}(R) = \{b \in B | (a, b) \in R\}$$

139. Define a binary relation  $R$  from  $A$  to  $B$  as follows:

$$R = \{(a, b) \in A \times B | a < b\}$$

140. Let  $A = \{1,2,3,4\}$  and define the following relations on  $A$  Then  $R = \{(1,3),(1,4),(2,3),(2,4),(3,1),(3,4)\}$  is \_\_\_\_\_.

$R$  is irreflexive

141. If  $A = \{1,2,3\}$  is a set and  $R = \{(2,3),(2,2),(3,4)\}$  is a relation on  $A$ , then  $R$  is

None of these

142.  $R$  is not transitive iff there are elements  $a, b, c$  in  $A$  such that

if  $(a, b) \in R$  and  $(b, c) \in R$  but  $(a, c) \notin R$

143.  $Y+7 = 10$  is a proposition?

False

144. Let  $P(n)$  be a propositional function defined for all positive integers  $n$ .  $P(n)$  is true for every positive integer  $n$  if

Principal of mathematical induction

145. Grass is green is a proposition.

True

146.  $4 + 2 = 6$  is a proposition.

True

147.  $4 + 2 = 7$  is a proposition.

True

148. There are four fingers in a hand is a proposition

True

149. Close the door is not a proposition.

True

150.  $x$  is greater than 2 is not a proposition.

True

151. He is very rich is not a proposition.

True

152. Operation of subtraction is a binary operation on the set of \_\_\_\_\_.

Integers

153. Let  $R$  be a relation on a set  $A$ . If  $R$  is symmetric then its complement is.

Symmetric relation

154. Let  $R$  be a relation on set  $A$ .  $R$  is reflexive if and only if, for all  $a \in A$ ,  $(a, a) \in R$  or

equivalently  $aRa$ . That is, each element of  $A$  is related to itself.

### Reflexive relation

155. Let  $R$  be a relation on a set  $A$ .  $R$  is transitive if and only if for all  $a, b, c \in A$ , if  $(a, b) \in R$  and  $(b, c) \in R$  then  $(a, c) \in R$ .

### Transitive relation

156. If  $p \leftrightarrow$  is true, then \_\_\_\_\_.

$P$  and  $q$  both are true

157. The 3<sup>rd</sup> term of the sequence  $b_n = 5^n$  is \_\_\_\_\_.

125

158. Which of the following biconditional statement is true \_\_\_\_\_.

$1+1 = 3$  if and only if earth is flat

159. Let  $A = \{0, 1, 2\}$  and  $R = \{(0, 2), (1, 1), (2, 0)\}$  be a relation on  $A$ . Then which of the following statement about  $R$  is true?

$R$  is symmetric

160. Let  $p$  be True and  $q$  be True, then  $(\sim p \text{ AND } q)$  is

False

161.  $9!/6! =$  \_\_\_\_\_.

504

162. Let  $f: X \rightarrow Y$ . The range of  $f$  consists of those elements of  $Y$  that are image of elements of  $X$

Range of a function

163. Let  $f$  be a function from  $X = \{2, 4, 5\}$  to  $Y = \{1, 2, 4, 6\}$  defined as;  $f = \{(2, 6), (4, 2), (5, 1)\}$  Then range of  $f$  is \_\_\_\_\_.

1, 2, 6

164. If  $A$  is defined as the set of English alphabets then the cardinality of  $A$  is \_\_\_\_\_.

26

165. If  $A$  is defined as the set of 467 people then the birthday of  $A$  is \_\_\_\_\_.

366 possible birthday  
166. . Among any group of 367 people, there must be at least two with the same birthday, because there are only

366 possible birthday

167. If  $(p \text{ XOR } q)$  is true, then \_\_\_\_\_.

P is false, q is true

168. The negation of the conditional statement ' if I work hard, then I will get A grade' is \_\_\_\_\_.

I do not work hard AND I will not get a grade mark

169. If  $p=\text{true}$  and  $q=\text{true}$  then the truth value of  $p \text{ XOR } q$  will be \_\_\_\_\_.

False

170. If  $p=\text{False}$  and  $q=\text{false}$  then the truth value of  $p \text{ XOR } q$  will be \_\_\_\_\_.

False

171. If  $p=\text{true}$  and  $q=\text{false}$  then the truth value of  $p \text{ XOR } q$  will be \_\_\_\_\_.

True

172. The set of natural number is equal to \_\_\_\_\_.

Set of positive integers

173. Let R and S be transitive relations on a set A the

Neither  $R \text{ UNION } S$  is transitive nor  $R \text{ intersection } S$  is transitive

174. If a relation  $R=\{(1,2)(2,3)(3,4)(4,1)(2,2)\}$  is given then which of the following is true about this Relation.

R is antisymmetric

175. Let R be a binary relation on a set A. R is anti-symmetric iff

$\forall a, b \in A$  if  $(a,b) \in R$  and  $(b,a) \in R$  then  $a = b$ .

176. A graph of a function f is onto iff every horizontal line intersects the graph in at least one point.

True

177. A graph of a function  $f$  is onto iff every horizontal line intersects the graph in at least one point.

Graph onto function

178. A graph of a function  $f$  is one-to-one iff every horizontal line intersects the graph in at most one point

GRAPH OF ONE-TO-ONE FUNCTION:

179. A graph of a function  $f$  is bijective iff every horizontal line intersects the graph at exactly one point.

Graph of bijective function

180.  $A = \{x: x \text{ is even number divisible by 6 and less than } 100\}$  is \_\_\_\_\_.

Set builder form

In venn diagram, sets are commonly represented by \_\_\_\_\_.

Circles and ovals....both

181. An infinite sequence may have only a finite number of values.

True

182. If  $n$  is even, then  $C_n = 2$  and if  $n$  is odd, then  $C_n = 0$ . Hence, the sequence oscillates endlessly between 2 and 0.

True

183. If  $n$  is a positive integer, then  $n! = n$  for  $n =$

1,2

184.  $n! = n(n-1)!$  for each

positive integer

185. A union  $A \cup A$ , A intersection  $A \cap A$  are called.

Commutative law.

186. '-' is a binary operation on the set of integers  $\mathbb{Z}$ .

True

187. "+" and "." are binary operations on the set of natural numbers  $\mathbb{N}$ .

True

188. “-” is not a binary operation on  $\mathbb{N}$ .

True

189. Let  $p = 1$  am a student, and  $q = I$  love to study mathematics, then the symbolic form for the

For the statement ‘I am not a student but I love to study’

$$\sim p \wedge q$$

190. A set is countably infinite if and only if, it has the same cardinality as the set ....

Positive integers

191. “ $\div$ ” is a binary operation on the set of non-zero rational numbers  $\mathbb{Q} - \{0\}$ , but not a binary operation on  $\mathbb{Z}$ .

True

192. If a relation  $R$  is reflexive, anti symmetric and transitive then which of the following is not true for the inverse relation.

Inverse relation will be irreflexive

193. An important data type in computer programming consists

Finite sequences

194. An important data type in computer programming consists of finite sequences known as

One-dimensional arrays

195. Negation of  $1 < x < 8$  is \_\_\_\_\_.

$$1 > x \text{ AND } x > 8$$

196. Which of the following is true about  $C_n = 1 - (-1)^n$  for all integers  $n > 0$

If  $n$  is even,  $C_n = 2$  and if  $n$  is odd, then  $C_n = 0$

197. For any set  $A$ , the Cartesian product of  $A$  and  $A$  is known as  
Universal relation

198. For any set  $A$ ,  $\emptyset$  is known as the

Empty relation

199. For any set  $A$ ,  $A \times A$  is known as the



## Universal relation

200. If  $F_k - 1 + F_{k-2}$  then  $F_0=1$ ,  $F_1=2$ , then  $F_2=$  \_\_\_\_\_.

3

201. In general  $n! = n(n-1)!$  For all integers  $n$

True

202. If  $f$  and  $g$  are two one-to-one functions, then their composition that is  $g \circ f$  is one-to-one

True

203. The converse of the conditional statement if I live in Quetta, then I live in Pakistan is \_\_\_\_\_.

IF I live in Pakistan, then I Live in Quetta

204. An integer  $n$  is a product square if and only if \_\_\_\_\_ for some integer  $k$

$N = k^2$

205. Inverse of a surjective function is always a function.

False

206. Inverse of a surjective function may not be a function.

True

207. If the conclusion is true when the premises are true, the argument will be \_\_\_\_\_.

Valid

208. if the conclusion is false when all the premises are true, the argument will be

invalid

209. The biconditional statement for the following statement would be \_\_\_\_\_.

The number is divisible by 6 and only if it is divisible by both 2 and 3

210. Universal bound law states that

$P \vee \neg P$

211.  $P$  is equivalent to  $q$  means \_\_\_\_\_.

**p is necessary and sufficient for q.**

212. The converse of the conditional statement, 'If I live in Lahore, then I live in Punjab' is \_\_\_\_\_.

If I live in Punjab, then I live in Lahore

213. Complementary relation symbolically written as \_\_\_\_\_.

$$R = A \times B - R = \{(a,b) \in A \times B \mid (a,b) \notin R\}$$

214.  $(s \vee \sim s)$  is the \_\_\_\_\_.

Tautology

215. The statement form  $p \wedge \sim p$  is a

Contradiction

216. Set difference law \_\_\_\_\_.

$$A - B = A \cap B^c$$

217. If A contains 3 elements and B contains 2 elements, then number of subsets of  $A * B$  are \_\_\_\_\_.

64

218. Neither Ali is a student, nor he is from Lahore. In the logical sentences, it can be written

As \_\_\_\_\_.

$$\sim p \wedge \sim q$$

219. If a is the 1<sup>st</sup> term and d be the common difference of an arithmetic sequence Then the sequence is  $a+a+d, a+2d, a+3d$ .

True

220. Let A and B be subset of universal set U then the difference of A and B or  $A - B =$

$$A \cup B = \{x \in U \mid x \in A \text{ or } x \in B\}$$

221. Let R be a binary relation on a set A. If R is anti symmetric then \_\_\_\_\_  
Inverse of R is anti symmetric

222. Let R be a binary relation on a set A. If R symmetric then

Inverse of R is symmetric

223. Let R be a binary relation on a set A. If R is reflexiv then

Inverse of R is reflexive

224. Let R be a binary relation on a set A. If R is anti transitive then

Inverse of R is transitive

225. Let  $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ , then which of the following sets represent the partition of the set A?

$A = \{1, 3, 5, 7, 9\}$ ,  $B = \{2, 4, 6\}$ ,  $C = \{8\}$

226. Let  $A = \{0, 1, 2\}$  and  $R = \{(0, 2), (1, 1), (2, 0)\}$  be a relation on A. Then which of the following ordered pairs

Are needed to make it transitive?

$(0, 0)$  and  $(2, 2)$

227. Elements of a set are listed in pair of brackets and separated by commas is \_\_\_\_\_.

Tabular form

228. A well defined collection of objects is called \_\_\_\_\_.

Set

229. Every set is not a proper subset of itself.

False

230. Every set is a subset of itself

True

231. Let A be subset of universal set U then double complements law is

$(A^c)^c = A$

232. The relation divides on the set of integers is \_\_\_\_\_.

An anti symmetric relation

233. For any set A,  $A \times A$  is known as

Universal set

234. For any set A,  $\emptyset$  is known as the

Empty relation

235. The logical statement  $p \wedge q$  means \_\_\_\_\_.

**P AND q**

236. Which of the logical expressions represents the sentence 'if you read stories, then your vocabulary will increase and conversely?'

**$p \rightarrow q$**

237. Vertical line test is used to determine that whether the graph of a relation is a Function or not.

**True**

238. The set Z all integers is countable

**True**

239. Number of one to one functions from  $X=\{a,b\}$  to  $Y=\{u,v\}$  are equal to

**2**

240.  $P \leftrightarrow q$  is true when \_\_\_\_\_.  
P is false, q is false

241. The set of all vowels in the english alphabet  
 $V=\{a,e,i,o,u\}$  is \_\_\_\_\_.

**Tabular and descriptive form....both**

242. Let R be a relation from a set A to a set B and S a relation from B to a set C. The composite of R and S denoted  $SoR$  is the relation from A to C, consisting of ordered pairs (a,c) where  $a \in A$ ,  $c \in C$ , and for which there exists an element  $b \in B$  such that  $(a,b) \in R$  and  $(b,c) \in S$ . Symbolically

**$SoR = \{(a,c) | a \in A, c \in C, \exists b \in B, (a,b) \in R \text{ and } (b,c) \in S\}$**

243. Let  $A=\{1,2,3\}$  and  $B=\{0,1,2\}$  and  $C=\{a,b\}$   
 $R=\{1,0\},\{1,2\},\{3,1\},\{3,2\}$   
 $S=\{0,b\},\{1,a\},\{2,b\}$   
Composite of R and S= \_\_\_\_\_.

**$\{1,a\},\{1,b\},\{2,a\},\{3,b\}$**

244.  $A \cap B =$

**An integer n is even if, and only if,  $n = 2k$  for some integer k.**

**True**

2. An integer n is odd if, and only if,  $n = 2k + 1$  for some integer k.

**True**

3. An integer n is prime if, and only if,  $n > 1$  and for all positive integers r and s, if  $n = r \cdot s$ , then  $r = 1$  or  $s = 1$ .

**True**

4. An integer  $n > 1$  is composite if, and only if,  $n = r \cdot s$  for some positive integers  $r$  and  $s$  with  $r \neq 1$  and  $s \neq 1$ .

True

5. A real number  $r$  is rational if, and only if,  $a/b$  for some integers  $a$  and  $b$  with  $b \neq 0$ .

True

6. If  $n$  and  $d$  are integers and  $d \neq 0$ , then  $d$  divides  $n$ , written  $d \mid n$  if, and only if,  $n = d \cdot k$  for some integers  $k$ .

True

7. An integer  $n$  is called a perfect square if, and only if,  $n = k^2$  for some integer  $k$ .

True

245.  $A \cap B =$

$$\{x \in U \mid x \in A \text{ and } x \in B\}$$

246. A set may be divided up into disjoint subsets, such division is called.

Partition

247. Let  $R$  be a binary relation defined on a set  $A$ .  $R$  is a partial order relation, if and only if,  $R$  is reflexive, antisymmetric, and transitive. The set  $A$  together with a partial ordering  $R$  is

Partially order relation

248. Let  $A$  and  $B$  be subsets of a universal set  $U$ . The difference of “ $A$  and  $B$ ” (or relative complement of  $B$  in  $A$ ) is the set of all elements in  $U$  that belong to  $A$  but not to  $B$ , and is denoted  $A - B$  or  $A \setminus B$ .

Difference

249. Let  $A$  and  $B$  be subsets of a universal set  $U$ . The difference of “ $A$  and  $B$ ” (or relative complement of  $B$  in  $A$ ) is the set of all elements in  $U$  that belong to  $A$  but not to  $B$ , and is denoted  $A - B$  or  $A \setminus B$ . symbolically

$$A - B = \{x \in U \mid x \in A \text{ and } x \notin B\}$$

250. Let  $A$  and  $B$  subsets of a universal set  $U$ . The intersection of sets  $A$  and  $B$  is the set of all elements in  $U$  that belong to both  $A$  and  $B$  and is denoted  $A \cap B$ .

Intersection

251. Let  $A$  and  $B$  subsets of a universal set  $U$ . The intersection of sets  $A$  and  $B$  is the set of all elements in  $U$  that belong to both  $A$  and  $B$  and is denoted  $A \cap B$ . symbolically

$$A \cap B = \{x \in U \mid x \in A \text{ and } x \in B\}$$

252. a set  $A$  is the set of all subsets of  $A$ , denoted  $P(A)$ .

power set

253. If A has n elements then P(A) has

$2^n$  elements

254. Every set can be represented in tabular form

True

255.  $\sim p$  is true when \_\_\_\_\_.

P is False

256. Negation of  $2 < x < 3$  is \_\_\_\_\_.

$2 > x$  Or  $x > 3$  (not sure)

257. The negation of an AND statement is logically equivalent to the OR statement in which each component is negated.

$$\sim (p \wedge q) \equiv \sim p \vee \sim q$$

258. The negation of an OR statement is logically equivalent to the AND statement in which each component is negated.

$$\sim (p \vee q) \equiv \sim p \wedge \sim q$$

259. A function  $f: \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = \text{square root}(x)$  is a real valued function

False

260.  $f: \mathbb{R}^+ \rightarrow \mathbb{R}$  defined by  $f(x) = \log x$  is a real valued function

true

261.  $g: \mathbb{R} \rightarrow \mathbb{R}$  defined by  $g(x) = e^x$  is a real valued function of a real variable.

True

262. Let f and g be real valued functions with the same domain X. That is  $f: X \rightarrow \mathbb{R}$  and  $g: X \rightarrow \mathbb{R}$ .

Sum of function

263. The sum of f and g denoted by  $f+g$  is a real valued function with the same domain X i.e.  $f+g: X \rightarrow \mathbb{R}$  defined by

$$(f+g)(x) = f(x) + g(x) \quad \forall x \in X$$

264. The matrix representation of an inverse relation can be obtained taking the \_\_\_\_\_ of the relational

## Matrix

### Inverse

265. Let  $R$  be a relation from  $A$  to  $B$ . The inverse relation  $R^{-1}$  from  $B$  to  $A$  is defined as:

### Inverse relation

266. Let  $R$  be a relation from  $A$  to  $B$ . The inverse relation  $R^{-1}$  from  $B$  to  $A$  is defined as:

$$R^{-1} = \{(b, a) \in B \times A \mid (a, b) \in R\}$$

267. In the directed graph of an antisymmetric relation there is \_\_\_\_\_ pair of arrows between two

Distinct elements of the set

No

268. For any set  $A$ , the Cartesian product of  $A$  and  $A$  is known as \_\_\_\_\_.

### Universal relation

269. A function whose range consists of only one element is called \_\_\_\_\_.

### Constant function

270. In the matrix representation of an irreflexive relation all the entries in the Main diagonal are \_\_\_\_\_.

0

271. A function whose inverse function exists is called an  
Invertible function

272. A sequence is a function whose domain is the set of integers greater than or equal to a particular integer  $n_0$

Formal

273. We use the notation  $a_n$  to denote the image of the integer  $n$ , and call it a term of the sequence

### Notation

274. If  $f$  and  $g$  are two one-to-one functions, then their composition  $g \circ f$  is  
One to one function

275. If  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$  are both onto functions, then  $g \circ f: X \rightarrow Z$  is

Onto

276. A graph of a function  $f$  is onto iff every horizontal line intersects the graph in at least

One point

277. Let A and B subset of universal set U then  $(A \cup B)^c =$

$$A^c \cap B^c$$

**CORRECT FILE**

1.  $X = a, X = 3, x = 5a$ -----

Arithmetic sequence

2. composition of a function is commutative operation False

3. real valued function is a function that assign----- to each number of its domain

real number

4. let X equal 1 2 3 4 and a function f depend on X by f into 1  
equal 1

f into 2 equals to 2

f into 3 equal 3 and

F into 4 equal 4

Then-----

F is an identity function



5. a constant function is Injective if and only if-----

the codomain consists of a single element

6. let  $f: \mathbb{R} \rightarrow \mathbb{R}$  is one to one function then  $c, f, c$  does not equal to zero is also, one to one function

True

7. if  $f$  and  $G$  are two one to one function then their composition that is  $g \circ f$  is one to one

True

8. let  $f(x) = 3x$  and  $g(x) = x + 2$  define function  $f$  and  $G$  from  $\mathbb{R}$  to  $\mathbb{R}$  then  $f \circ g(x)$  is-----

$$\underline{3x^2 + 6x}$$

9. the set  $\mathbb{Z}$  of all integers is countable

True

10.  $0.818181$  is an infinite geometric series

True

11. which of the following is not correct for a sequence?

A sequence is a relation whose domain is the set of natural numbers

12.  $f(x) = x^2$  is not one to one function from  $\mathbb{R}$  to  $\mathbb{R}$  True

13. let  $f: \mathbb{R} \rightarrow \mathbb{R}$  equal  $f(x) = x + 2$  then  $f^{-1}(x)$  is

$$\underline{x - 2}$$

14. let  $f: \mathbb{R} \rightarrow \mathbb{R}$  equal  $f(x) = x^2 + 1$  define function  $f$  from  $\mathbb{R}$  to  $\mathbb{R}$  and  $C = 2$  to be any scalar then  $c, f(x)$  is-----

$$\underline{2x^2 + 2}$$

15. one to one correspondence means the condition of both a  
and c

16. Term of a geometric sequence is 2 and common ratio is  $\frac{1}{2}$  then third term of a sequence is

$$\underline{\frac{1}{2}}$$

17. the process of defining an object in terms of smaller version of itself is called recursion

True

18. a set is called countable if and only if it is

(d) (b) (c)

19. A sequence whose terms alternate in sign is called an----

alternating sequence

20. let  $f$  and  $G$  be two functions Defined by  $f(x) = x + 2$  and  $g(x) = x + 1$  then the composition of  $f$  and  $G$  is-----

$2x + 3$

21. number of one-to-one functions from  $X = \{a, b\}$  to  $Y = \{u, v\}$  are equal to-----

2

22. an important data type in computer programming consists of -----

finite sequence

23. if a function is recursively defined as  $F(0) = 4$  &  $f(n + 1) = 3f(n)$  then  $f(1) =$  -----

15

24. inverse of a function may not be a function

True

25. if a function is recursively defined as  $F(0) = -1$  &  $f(n + 1) = f(n) + 3$ , then  $f(1) =$  -----

2

26. The function  $f$  and  $G$  are inverse of each other if and only if their composition gives-----

identity function

27. which of the following state is a domain of a sequence ----

set of natural number

28. Let  $f$  and  $g$  be two functions defined by  $f(x) = x + 2$  and  $g(x) = 2x + 1$ . Then the composition of  $f$  and  $g$

$2x + 5$

29. Let  $R = \{(1,2), (3,4), (5,6), (7,8)\}$ . Range of the inverse of the relation is -

-----

$\{1,3,5,7\}$

30. The set  $Z$  of all integers is countable.

True

31. In the representation of an irreflexive relation using a matrix, all its diagonal entries will be -----

0

32. Let  $p =$  I am hard working.  $Q =$  I am intelligent. Then  $p \wedge q$  means ----

----

I am NOT hard working BUT I am intelligent.

33. If a set A contains n elements then the number of elements in its power set P(A) is-----

$$\underline{2^n}$$

34. Which of the following is true for the following sequence?  $C_n = 1 - (-1)^n$  for all integers  $n > 0$

If n is even, then  $C_n = 0$  and if n odd, then  $C_n = 2$

35. Let  $A = \{1, 2, 3, 4\}$  and  $R = \{(1, 1) (2, 2) (3, 3) (4, 4)\}$  then R is

Options a, b and c all true

36. If  $A - B = A$ , which of the following statement is true? All of the given statements are false

37. If  $A = \{1, 2, 3\}$  is a set and  $R = \{(1, 2) (2, 2) (2, 1)\}$  is a relation on A. R is Symmetric

38. The converse of the conditional statement 'If I live in Lahore, then I live in Punjab' is -----

If I live in Punjab, then I live in Lahore

39. Every set can be represented in tabular form.

False

40. The function defined from  $\mathbb{Z}$  to  $\mathbb{Z}$  as  $f(x) = 1 / (x + 2)(x - 2)$  is not well defined because-----

Function is not defined at  $x = -2$  and  $x = 2$

41. Which of the following is always true for the matrix representation of asymmetric relation?

Matrix is equal to its transpose

42. Associative law of union for three sets is-----

$A \cup (B \cup C) = (A \cup B) \cup C$

43. If  $f$  is bijective function then  $(f^{-1}(f(x)))$  is equal to ----

$x$

44.  $P \leftrightarrow Q$  is logically equivalent to -----

$(p \rightarrow q)$

45. Let  $A = \{0, 1, 2\}$  and  $R = \{(0, 2), (1, 1), (2, 0)\}$  be a relation on  $A$ . Then which of the following ordered pairs are needed to make it reflexive.

$(0, 0)$  and  $(2, 2)$

46. Let  $A$  be subset of universal set  $U$  then  $A \cup A^c = \underline{\Phi}$

47. Let  $R$  be a binary relation on a set  $A$ . if  $R$  is anti-symmetric then -----

Inverse of  $R$  is anti-symmetric

48. The inverse of statement 'if it is pleasant outside, then I go to the park' is -----

If it is NOT pleasant outside, then I do NOT go to the park.

49. The inverse of the statement 'if it is hot day, then I am at home' is -----

---

if it is NOT hot day, then I am NOT at home

50. Let  $A, B, C$  be subset of universal set  $U$  then the idempotent law is -----

$A \cap A = A$

51. Let A and B be subset of universal set U then  $A - (A - B) = \underline{A \cap B}$

52. Real valued function is a function that assigns -----to each member of its domain.

Only a real number

53. Let R be a relation on set A, if R is reflexive then its complement is -----

Irreflexive

54. Range of function  $f(x) = \{e^x\}$  is -----

Set of positive Real numbers

55. Composite relation symbolically written as -----

$SoR = \{(a,c) | a \in A, c \in C, \exists b \in B, (a,b) \in R \text{ and } (b,c) \in S\}$

56. If  $x \equiv 17 \pmod{5}$  which of the following integers are valid solution for x? 12

57. Operation of subtraction is a binary operation on the set of -----

Integers

58. Range of relation  $\{(0, 10), (3, 22), (90, 34)\}$  is -----

$\{10, 22, 34\}$



59. Let  $S = R$  and the “square” relation  $R = \{(x, y) | x^2 = y^2\}$ . The square relation is an ----- relation.

Equivalence relation

60. The logic gate NOT is a unary operation on  $\{0, 1\}$  True

61. Let  $A = \{1, 2\}$ , then  $P(A) = \text{-----}$

$\{\{\}, \{1\}, \{2\}, \{1, 2\}\}$

62. If a relation  $R$  is reflexive, anti symmetric and transitive then which of the following is not true for the inverse relation.

Inverse relation will be irreflexive.

63. Let  $R$  be a binary relation on a set  $A$ .  $R$  is anti-symmetric iff -----

$a, b \in A$  if  $(a, b) \in R$  and  $(b, a) \in R$  then  $a = b$

64. the inverse relation  $R^{-1}$  from  $B$  to  $A$  is defined as -----

$R^{-1} = \{(b, a) \in B \times A \mid (a, b) \in R\}$

65. Which of the following is true for the matrix representation of asymmetric relation?

Matrix is equal to its transpose

66. If  $x \equiv -10 \pmod{15}$  which of the following integers are valid solutions for  $x$ ?  
5

67. Let  $A = \{0, 1\}$  and  $B = \{1\}$ . Let  $R$  and  $S$  be two binary relations on the Cartesian product of  $A$  and  $B$  such that  $R = \{(0, 1)\}$  and  $S = \{(1, 1)\}$

Then  $R \cap S =$  -----

Empty

68. The domain of a relation  $R$  from  $A$  to  $B$  is symbolically written as:  $D(R) = \{a \in A \mid \exists b \in B \text{ such that } (a, b) \in R\}$

69. Which of the following is not a type of a relation?

Permutation

70. Let  $A = \{4, 5, 6\}$  then the relation  $R = \{(4, 5), (5, 4), (6, 5), (5, 6)\}$  is -----  
-----

$R$  is symmetric

71. Let  $A = \{1, 2, 3\}$  and  $B = \{0, 1, 2\}$  and  $C = \{a, b\}$

$R = \{(1, 0), (1, 2), (3, 1), (3, 2)\}$

$S = \{(0, b), (1, a), (2, b)\}$

Composite of  $R$  and  $S =$  -----

$\{(1, b), (3, a), (3, b)\}$

72. If A contain 3 elements and B contain 2 elements, then number of subsets of  $A \times B$  are -----

64

73. Let  $A = \{1,2,3,4\}$  and define the following relations on A then  $R = \{(1,3)(1,4) (2,3 (2,4) (3,1) (3,4)\}$  is -----

R is irreflexive

74. Let  $X = \{2,4,5\}$  and  $Y = \{1,2,3\}$  and R be a relation from X to Y defined by  $R = \{(2,4) (4,1) (a,2)\}$ . for what value of 'a' the relation R is a function?

5

75. A function whose range consists of only one element is called \_\_\_\_\_

One to one function

76. The set Z of all integers is \_\_\_\_\_

Countable

77. One-to-one correspondence means the condition of \_\_\_\_\_

Both (a) and (c)

78. Let  $X = \{1,5,9\}$  and  $Y = \{3,4,7\}$ , Define a function f from X to Y such that

$f(1)=7, f(5)=3, f(9)=$  \_\_\_\_\_. Which is true f(9) to make it a one-to-one (injective) function?

4

79. What will be the fourth term of the following sequence  $1/2, 2/3, 3/4$   
\_\_\_\_\_?

$4/5$

80. The value of  $6!$  =

720

81. A constant function is one to one iff its \_\_\_\_\_ is a singleton.

Domain

82. A constant function is onto iff its \_\_\_\_\_ is a singleton. Co-domain

Number of one to one functions from  $X=\{a,b\}$  to  $Y=\{u,v\}$  are equal to \_\_\_\_\_ -

2

83. If  $f$  is defined recursively by  $f(0) = -1$  and  $f(n+1) = f(n) + 3$ , then  $f(2) =$  \_\_\_\_.

5

84. A function whose inverse function exists is called a/an \_\_\_\_\_

Invertible function

85. Let  $f(2)=3$ ,  $g(2)=3$ ,  $f(4)=1$  and  $g(4)=2$  then the value of  $f \circ g(4)$  is.....3

86. Let  $A=\{1,2,3,4\}$  and  $B=\{7\}$  then the constant function from A to B is

\_\_\_\_\_.

Onto

87. Composition of a function is a commutative operation.

False

88. Composition of a function is not a commutative operation True

89. The sum of first five whole number is\_\_\_\_\_.

10

90. If f and g are two one-to-one functions, then their composition that is  $g \circ f$  is one-to-one.

True

91. Inverse of a surjective function is always a function.

False

92. Inverse of a surjective function may not be a function True

93. Let  $X=\{1,2,3,4\}$  and  $Y=\{7,8,9\}$  and let f be function defined from X to Y such that f is onto then which of the following statement about f is true?

Co-domain of f must contain 3 elements

94. If  $f; X \rightarrow Y$  and  $g; Y \rightarrow Z$  are both onto functions. Then  $g \circ f; X \rightarrow Z$  is

\_\_\_\_\_

onto

95. If  $f$  and  $g$  are two one-to-one functions, then their composition  $g \circ f$  is\_\_

One-to-one

96. If  $f: W \rightarrow X$ ,  $g: X \rightarrow Y$ , and  $h: Y \rightarrow Z$  are functions, then \_\_\_\_\_  
(hog)of = ho(gof)

97. Cardinality of positive prime numbers less than 20 is \_\_\_\_\_.

8

98. If  $f(x) = \sin^{-1}(x)$  and  $g(x) = \sin x$  then  $g \circ f(x)$  is \_\_\_\_\_.

X

99.  $0! =$  \_\_\_\_\_.

1

100. An important data type in computer programming consists of

\_\_\_\_\_.

Finite sequences

101. Let  $f(x) = 2x$  and  $g(x) = x + 2$  define functions  $f$  and  $g$  from  $R$  to  $R$ , then  $(f \circ g)(x)$  is

\_\_\_\_\_.

x-2

102. The total number of terms in an arithmetic series  
 $0+5+10+15+....50$  are\_\_\_\_\_.

11

103.  $9!/6!=$ \_\_\_\_\_

504

104. Let  $f(x)=3x$  and  $g(x)=3x-2$  define functions  $f$  and  $g$  from  $\mathbb{R}$  to  $\mathbb{R}$ , THEN  
 $(F+G)(X)$  is

\_\_\_\_\_

$6x-2$

105. If  $f$  is a bijective function then  $(f-1f(x))$  is equal to

X

106. A sequence whose terms alternate in sign is called an\_\_\_\_\_

Alternating sequence

107. Common ration in the sequence “4, 16, 64, 256,...” is.....12.

4

108. 0.8181818181 is a infinite geometric series.True

109. The word 'algorithm' refers to a step-by-step method for performing some action.

True

110. A predicate become \_\_\_\_\_ when its variables are given specific values

Sentence

111. The sum of two irrational numbers must be an irrational number. False

112. The sum of two irrational numbers need not be irrational number True

113. The division by zero is allowed in mathematics. False

114. The product of any two consecutive positive integers is divisible by 2

True

115. If 'n' is an odd integer then  $n^3 + n$  is \_\_\_\_\_.

Even

116. For integers a, b, c, If a divides b and a divides c, then a divides (a+b).



**False**

**117. Quotient remainder theorem states that for any positive integer  $d$ , there exist unique integer  $q$  and  $r$  such that \_\_\_\_\_ and  $0 < r < d$**

$$N = d \cdot q + r$$

**118. A rule that assigns a numerical value to each outcome in a sample space is called**

**Random variable**

**119. If  $A$  and  $B$  are two disjoint (mutually exclusive) events then  $P(A \cap B) =$**

$$P(A) + P(B)$$

**120. How many ways are there to select five players from a 10 member tennis team to make a trip to a match to another school?**

$$C(10, 5)$$

**The expectation of  $x$  is equal to**

$$\sum x f(x)$$

**121. If  $P(A \cap B) = P(A) P(B)$  THEN THE events  $A$  and  $B$  are called**

**Independent**

**122. A walk that starts and ends at the same vertex is called.**

None optins

123. How many integers from 1 through 1000 are neither multiple of 3 nor multiple of 5

497

124. What is the probability of getting a number greater than 4 when adie is thrown?

3/5

125. Eater formula for graphs is\_\_\_\_\_.

$F=e-v+2$

126.  $X+a, x+3a, x+5a, \dots$  is an \_\_\_\_\_

Arithmetic sequence

127. Composition of a function is a commutative operation.False

128. Real valued function is a function that assigns\_\_\_\_\_to each member of its domain.

Only a real number

129. Let  $X=\{1,2,3,4\}$  and a function 'f' defined on X  
 $f(1)=1, f(2)=2, f(3)=3, f(4)=4$  then \_\_\_\_\_

F is an identity function

130. A constant function is surjective if and only if \_\_\_\_\_  
The co-domain consists of a single element

131. Cardinality means the total number of elements in a set. True

132. If  $f(x)=2x$  and  $g(x)=x$  then  $g(f(x))$  is \_\_\_\_\_.  
 $2x^2$

133. Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  is one to one function then  $c, f, c$  is not equal 0 is also one to one function.  
True

134. Let  $X=\{1,2,3,4\}$  and a function 'f' defined from X to X by  $f(1)=1, f(2)=1, f(3)=1, f(4)=1$   
then which of the following is true? F is a  
constant function

135. If f and g are two one-to-one functions, then their composition that is  $g \circ f$  is one-to-one.  
True

136. Which of the following is not correct for a 'sequence'? A  
sequence is a relation whose domain is the set of natural numbers

137.  $F(x)=x^2$  is not one to one function from  $\mathbb{R}$  to  $\mathbb{R}^+$

True

138. Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  is one to one function then  $c, f \circ c$  is not equal to  $c$  is also one to one function.

True

139. Let  $f(x) = x + 2$  then  $f^{-1}(x)$  is \_\_\_\_\_  
 $x - 2$

140. Let  $f(x) = x^2 + 1$  define functions  $f$  from  $\mathbb{R}$  to  $\mathbb{R}$  and  $c = 2$  be any scalar, then  $c, f(x)$  is \_\_\_\_\_.  
 $2x^2$

141. One to one correspondence means the condition of \_\_\_\_\_.  
Both (a) and (c)

142. A function  $F: \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = \text{square root } x$  is a real valued function.  
False

143. If  $g: \mathbb{R} \rightarrow \mathbb{R}$  defined by  $g(x) = e^x$  is a real valued function of a real variable. True

144. A function  $F: \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = \log x$  is a real valued function

True

145.  $\frac{1}{2}$ , then 3rd term of sequence is \_\_\_\_\_.

$\frac{1}{2}$

146. The process of defining an object in terms of smaller versions of itself is called recursion.

True

147. Which of the following is not correct for a 'sequence'? A sequence is a relation whose domain is the set of natural numbers

148. A set is called countable if, and only if, it is \_\_\_\_\_.  
Finite and countable infinite.....  
both

149. A set that is not countable is called \_\_\_\_\_.  
Uncountable

150. A sequence whose terms alternate in sign is called an \_\_\_\_\_.  
Alternating sequence

151. Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  is one to one function then  $c \circ f$  is also one to one function for \_\_\_\_.

C is not equal 0

152. Let  $f(x) = x+3$  then  $f^{-1}(x)$  is \_\_\_\_\_ -  
 $x-3$

153. Let  $f$  and  $g$  be two functions defined by  $f(x) = x+2$  and  $g(x) = 2x+1$ .  
Then the  
composition of  $f$  and  $g$  is \_\_\_\_\_.  
 $2x+3$

154. Number of one to one functions form  $X=\{a,b\}$  to  $Y=\{u,v\}$  are equal to \_  
\_\_\_\_\_  
2

155. The fibonacci sequence is deined as  $F_0=1, F_1=1, F_k=F_{k-1}+F_{k-2}$  for all  
integers  $k > 2$  then which of the following is true for  $F_2$   
 $F_2-F_1 = 2+1=3$

156.  $x+a, x+3a, x+5, a, \dots$  is a/an \_\_\_\_\_.  
Arithmetic sequence

157. Inverse of a function may not be a function. True

158. In the following sequence  $a_k = K/(k+1)$ , for  $k=1$ ,  $a_1$  will be  
\_\_\_\_\_.  
 $\frac{1}{2}$

**159.** If  $f(x)=x$  and  $g(x)=-x$  are both one to one function then  $(f+g)(x)$  is also one to one function.

False

**160.** If  $f(x)=x$  and  $g(x)=-x$  are both one to one function then  $(f+g)(x)$  is not one to one function.

True

**161.** The function 'f' and 'g' are inverse of each other if and only if their composition gives \_\_\_\_\_.

Identity function

**162.** Which of the following set is the domain of a sequence? Set of real numbers

**163.** Let C is defined as the set of all countries in the world then C is a \_\_\_\_\_.

Finite set

**164.** A constant function is surjective if and only if \_\_\_\_\_.  
The co domain consists of a single element

**165.** Inverse of a function may not be a function. True

166. Let  $X=\{1,2,3,4\}$  and a function 'f' defined from X to X by  $f(1)=1, f(2)=1, f(3)=1, f(4)=1$  then which of the following is true?

F is a constant function

167. The composition of function is always  
Associative

168. A set is countably infinite if, and only if, it has the same  
cardinality as the set of -----  
Positive integers

169. Two functions 'f' and 'g' from 'X' to 'Y' are said to be equal if and only  
if \_\_\_\_\_.  
 $F(x)=g(x)$  for all 'x' belongs to X

170.  $Y=x^3$  is a graph of bijective function from R to R. True

171. Domain and range are same for \_\_\_\_\_.  
Identity function

172. Let  $X=\{1,2,3,4\}$  and a function 'f' defined on X by  
 $f(1)=1, f(2)=2, f(3)=3$   
F is an identity function

173. Composition of a function is a commutative operation. True



174. Inverse of a surjective function is always a function. False

175. A function whose inverse function exists is called a/an \_\_\_\_\_.

Invertible

176. Given a set  $X$  define a function  $I$  from  $X$  to  $X$  by  $i(x)=x$  from all  $x$  belonging to  $X$ . Then \_\_\_\_\_.

$I$  is both injective and surjective

177. Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  is a one to one function then  $c, f$  is also one to one function for.

$C$  is not equal 0

178. Let  $X=\{1,5,9\}$  and  $Y=\{3,4,7\}$ . Define a function  $f$  from  $X$  to  $Y$  such that  $f(1)=7, f(5)=3, f(9)=$  \_\_\_\_\_. Which is true for  $f(9)$  to make it a one-to-one (injective) function?

4

179. Which of the following is not a predecessors of  $a_k$ ?  $A_{k+1}$

180. Two functions ' $f$ ' and ' $g$ ' from  $X$  to  $Y$  are said to be equal if and only if \_\_\_\_\_.

$F(x)=g(x)$  for all 'x' belongs to x

181. The two functions 'f' and 'g' are equal if \_\_\_\_\_.

$F(x) = 3x$  and  $g(x) = 6x^2 + 3x/2x^2 + 1$  for all  $x \in \mathbb{R}$

182. If first term of a geometric sequence is 2 and common ratio is  $\frac{1}{2}$ , then-----  
3rd ter, of sequence.  $\frac{1}{4}$

183.  $Y = \text{squire root } x$  is an \_\_\_\_\_ function form  $\mathbb{R}^+$  to  $\mathbb{R}$   
One to one function

184.  $Y = x^2$  is an \_\_\_\_\_ function form  $\mathbb{R}$  to  $\mathbb{R}^+$   
NOT ONE TO ONE FUNTION

185. If a function  $(g \circ f)(x) : X \rightarrow Z$  is defined as  $(g \circ f)(x) = g(f(x))$  for all  $x \in X$ , Then  
the  
function -----.  
(gof)

186. If 0 is the first term and -2 be the common difference of an arithmetic  
series, then the sum of first five terms of series is \_\_\_\_\_.  
-20

187. If  $f(x) = \sin^{-1}(x)$  and  $g(x) = \sin x$  then  $g \circ f(x)$  is \_\_\_\_\_.  
X

188.  $f: X \rightarrow Y$  that is both one to one and onto is called a\_\_\_\_\_.

**Bijjective function**

189. What does 'y' denotes in a geometric sequence?

**Common ratio**

190. Let  $g$  be a function defined by  $g(x)=x+1$ . Then the composition of  $(g \circ g)$ .  $X$   
 $2+2x+1$

191. A graph of a function  $f$  is one to one iff every horizontal line intersects the graph in at most one point.

**True**

192. Which of the following is true for the following sequence? If  $n$  is even, then  $C_n=2$  and if  $n$  is odd, then  $C_n = 0$

193. The function ' $f$ ' and ' $g$ ' are inverse of each other if and only if their composition gives\_\_\_\_\_.

**Identity function**

194.  $N!$  is defined to be\_\_\_\_\_.

**The product of the integers from 1 to  $n$**

195. Let  $A = \{1,2,3,4\}$  and  $B=\{7\}$  then the constant function from A to B  
Both one to one and onto

196. A set is called countable if, and only if, it is \_\_\_\_\_.  
Countably infinite and finite

197. If  $f(x) = x$  and  $g(x) = -x$  are both one to one functions then  $(f+g)(x)$  is also one to one function.  
False

198. If  $a$  is the 1st term and  $d$  be the common difference of an arithmetic sequence  
then the sequence is  $a, a+d, a+2d, a+3d, \dots$ . True

199.  $x+a, x+3a, x+5, \dots$  is  $a/a_n$  \_\_\_\_\_.  
Arithmetic sequence

200. inverse of an injective function may not be a function. True

201.  $y=x^3$  is a graph of bijective function from  $\mathbb{R}$  to  $\mathbb{R}$ . False

202. Two functions 'f' and 'g' from x to y are said to be equal if and only if,  
 $f(x)=g(x)$  for all 'x' belongs to x

203. Common ratio in sequence '36, 12, 4, 4/3, .....' is .....1/3

204. A set is called finite if, and only if, is the \_\_\_\_\_ or there is -----  
-.

Empty set or one-to-one

205. Let  $f(x)=x^2-1$  and  $g(x)=x+1$  define functions f and g from R to R, then  $(f/g)x$   
 $x-1$

206. A graph of a function f is one to one iff every horizontal line intersects  
the  
graph in at most one point.

True

207. Let g be a function defined by  $g(x) = x+1$ . Then the composition of (gog).  
 $x^2+2x+1$

208.  $f:X \rightarrow Y$  that is both one to one and onto is called a \_\_\_\_\_.  
Bijective function

209. What does 'r' denotes in a geometric sequence?

Common ratio

210. Let  $A=\{1,2,3,4\}$  and  $B=\{7\}$  then the constant function from A to B is \_\_\_\_.

Both one to one and onto

211.  $n!$  is defined to be \_\_\_\_\_.

The product of the integers from 1 to n

212. A set is called countable if, and only if, it is \_\_\_\_\_. Both b and c

213. Which of the following is the example of an alternating sequence?  $C_n = n/n+1$  for  $n > 0$

214  $x+a, x+3a, x+5a, \dots$  is an \_\_\_\_\_

Arithmetic sequence

215  $0, -5, -10, -15, \dots$  is an \_\_\_\_\_.

Arithmetic sequence

225.  $5, 9, 13, 17, \dots$  is an \_\_\_\_\_.

Arithmetic sequence

226. An important data type in computer programming consists of  
\_. Finite sequence

227. One-to-one correspondence means the condition of\_\_\_\_\_.

One-one and onto ....both

228. Cardinality means the total number of elements in a set.True

229. Inverse of a function may not be a function.True

230. The functions 'f' and 'g' are inverse of each other if and only if their  
composition gives\_\_\_\_\_.

Identity function

231. A set is called finite if , and only if, it is the\_\_\_\_\_or there is  
\_\_\_\_\_.

Empty set or one-to-one

232. Let f and g be the two functions from R to R defined by  $f(x) = |x|$  and  $g(x) =$   
square root  $x^2$  for all  $x \in R$ , then\_\_\_\_\_.

F(x) is not equal to g(x)

234. If  $f^{-1}(x) = 6 - x/2$  then  $f^{-1}(2)$  is \_\_\_\_\_.

2

235. Let  $f(2)=3$ ,  $g(2)=3$ ,  $f(4)=1$  and  $g(4)=2$  then the value of  $f \circ g(4)$  is \_\_\_\_\_.

3

236. An important data type in computer programming consists of \_\_\_\_\_.

Finite sequence

237. Let  $f(x)=x+3$  then  $f^{-1}$

(x) is \_\_\_\_\_.

$x-3$

238. If  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$  are both onto function. Then  $g \circ f: X \rightarrow Z$  is

\_\_\_\_\_.

One-to-one function

239. The functions 'f' and 'g' are inverse of each other if and only if their composition gives

Identity function

240. The two function 'f' and 'g' are equal if \_\_\_\_\_.

$f(x) = 3x$  and  $g(x) = 6x^2 + 3x/2x + 1$  for all  $x \in \mathbb{R}$

241, Two functions 'f' and 'g' from X To Y are said to be equal if and only if--

----.



$F(x)$  and  $g(x)$  for all 'x' belongs to X

242 Which of the following set is the domain of a sequence? Set of natural numbers

243 If 1st term of a geometric sequence is 2 and common ratio is  $\frac{1}{2}$ , then 3<sup>rd</sup> term of sequence is\_\_\_\_\_.

$\frac{1}{2}$

244 Composition of a function is a commutative operation. True

245 If a function  $(g \circ f)(x): X \rightarrow Z$  is defined as  $(g \circ f)(x) = g(f(x))$  for all  $x \in X$ . Then the

function\_\_\_\_\_is known as composition of f and g.  $(g \circ f)$

246 A set is countably infinite if and only if, it has the same cardinality as the set of\_\_\_\_\_.

Positive integers

247 The 3rd term of the sequence  $b_n = 5n$  is\_\_\_\_\_.

125

248 If f is a bijective function then  $(f^{-1})(f(x))$  is equal to\_\_\_\_\_.

x

249 Let  $f(x) = x$  and  $g(x) = -x$  for all  $x \in \mathbb{R}$ , then  $(f+g)(x)$  is \_\_\_\_\_.

0

250 An infinite sequence may have only a finite number of values. True

251 The functions  $f \circ g$  and  $g \circ f$  are always equal. False

252 If  $f$  and  $g$  are two one-to-one functions, then their composition that is  $g \circ f$  is one-to-one.

True

253 A function whose range consists of only one element is called  
.. Constant function

254 Let  $X = \{1, 5, 9\}$  and  $Y = \{3, 4, 7\}$ . Define a function  $f$  from  $X$  to  $Y$  such that  $f(1)=7$ ,  $f(5)=3$ ,  $f(9)=4$  then which of the following statement about 'f' is true? F is both one-to-one and onto

255  $Y=x^3$  is a graph of bijective function from  $\mathbb{R}$  to  $\mathbb{R}$ . True

256 A function whose inverse function exists is called a/an\_\_\_\_\_.

Invertible

257 Let  $f$  and  $g$  be two functions defined by  $f(x) = x+2$  and  $g(x) = 2x+1$ . Then the composition of  $f$  and  $g$  is\_\_\_\_\_.

$2x + 5$

257 If  $r$  is a positive real number, then the value of  $r$  in  $3, r, r = -27r$  is None options

258 The\_\_\_\_\_of the terms of a sequence forms a series.Sum

259 The sum of first five whole number is\_\_\_\_\_.

10

260 If  $f_k = f_{k-1} + f_{k-2}$  then  $f_0 = 1, f_1 = 2$ , then  $f_2 =$ \_\_\_\_\_.

3

261 Range of function  $f(x) = \{e^x\}$  is -----

Set of positive Real numbers

262 Composite relation symbolically written as -----

$SoR = \{(a,c) | a \in A, c \in C, \exists b \in B, (A,B) \in R \text{ and } (b,c) \in S\}$



Q63 If  $x \equiv 17 \pmod{5}$  which of the following integers are valid solution for x?12

Q64 Operation of subtraction is a binary operation on the set of -----  
integers

Q65 Range of relation  $\{(0, 10), (3, 22), (90, 34)\}$  is -----  
 $\{1, 22, 34\}$

Remember own prayer

## Quiz 1 updated 2021

1. Range of function  $f(x) = (e^x)$  is \_\_\_\_\_

Set of positive real numbers

2. Composite relation symbolically written as \_\_\_\_\_

$S \circ R = \{(a,c) \mid a \in A, c \in C, \exists b \in B, (a,b) \in R \text{ and } (b,c) \in S\}$

3. If  $x \equiv 17 \pmod{5}$  which of the following integers are valid solution for  $x$  ?

12

4. Range of the relation  $\{(0,1), (3,22), (90,34)\}$

$\{1,22,34\}$

5. Let  $A = \{0,1,2\}$  and  $R = \{(0,2), (1,1), (2,0)\}$  be a relation on  $A$ . The which of the following ordered pairs are needed to make it transitive?

$(0,0)$  and  $(2,2)$

6. Operation of subtraction is a binary operation on the set of \_\_\_\_\_

Integers

7. Let  $S = \mathbb{R}$  and define the 'square' relation  $R = \{(x,y) \mid x^2 = y^2\}$ . The square relation is an \_\_\_\_\_ relation

Equivalence relation

8. The logic gate NOT is a unary operation on  $\{0,1\}$ .

True

9. Let  $A = \{1,2\}$ , then  $P(A) =$  \_\_\_\_\_

$\{\emptyset, \{1\}, \{2\}, \{1,2\}\}$

10. If a relation  $R$  is reflexive, anti symmetric and transitive then which of the following is not true for the inverse relation.

Inverse relation will be irreflexive.

11. Let  $R$  be a binary relation on a set  $A$ ,  $R$  is anti-symmetric iff \_\_\_\_\_

$a, b \in A$  if  $(a,b) \in R$  and  $(b,a) \in R$  then  $a=b$

12. “-“ is a binary operation on the set of integers  $\mathbb{Z}$ .

**True**

13. The inverse relation  $R^{-1}$  from  $B$  to  $A$  is defined as \_\_\_\_\_

**$R^{-1} = \{(b,a) \in B \times A \mid (a,b) \in R\}$**

14. Which of the following is always true for the matrix representation of a symmetric relation?

**Matrix is equal to its transpose**

15. Let  $A = \{1,2,3,4\}$  and let  $R$  and  $S$  be transitive binary relations on  $A$  defined as;  $R = \{(1,2), (1,3), (2,2), (3,3), (4,2), (4,3)\}$  and  $S = \{(2,1), (2,4), (3,3)\}$  then  $R \cup S = \{(1,2), (1,3), (2,1), (2,2), (2,4), (3,3), (4,2), (4,3)\}$

**$R \cup S$  is transitive**

16. Let  $S = R$  and define the ‘square’ relation  $R = \{(x,y) \mid x^2 = y^2\}$ . The square is an \_\_\_\_\_ relation.

**Equivalence relation**

17. If  $x \equiv -10 \pmod{15}$ . Which of the following integers are valid solution for  $x$ ?

**5**

18. Let  $R$  be a binary relation on a set  $A$ . If  $R$  is anti symmetric then \_\_\_\_\_

**Inverse of  $R$  is anti symmetric**

19. If  $A = \{1,2,3\}$  is a set and  $R = \{(1,2), (2,2), (2,1)\}$  is a relation on  $A$ ,  $R$  is

**Symmetric**

20. Let  $A = \{0,1\}$  and  $B = \{1\}$ . Let  $R$  and  $S$  be two binary relations on Cartesian product of  $A$  and  $B$  such that  $R = \{(0,1)\}$  and  $S = \{(1,1)\}$ . Then  $R \cap S =$  \_\_\_\_\_

**Empty**

21. A relation  $R$  is said to be \_\_\_\_\_ iff it is reflexive, antisymmetric and transitive.

**Partial order Relation**

22. Let  $X = \{1,2,3\}$  and  $Y = \{7,8,9\}$  and let  $f$  be function defined from  $X$  to  $Y$  such that  $f$  is onto then which of the following statement about  $f$  is true?

**Co-domain of  $f$  must contain 1 element**

23. The function  $f \circ g$  and  $g \circ f$  are always equal

**False**

24. If a relation  $R = \{(1,2), (2,3), (3,4), (4,1), (2,2)\}$  is given then which of the following is true about this relation

**$R$  is reflexive**

25. A set is called countable if, and only if, it is \_\_\_\_\_

**(a) finite**

26. Let  $f(x) = x^2 - 1$  define function  $f$  from  $\mathbb{R}$  to  $\mathbb{R}$  and  $c = 2$  be any scalar, then  $c, f(x)$  is \_\_\_\_\_

**$2x^2 - 2$**

27. The set  $\mathbb{Z}$  of all integers is \_\_\_\_\_

**Countable**

28. Let  $R$  be a binary relation on a set  $A$ . If  $R$  is anti symmetric then \_\_\_\_\_

**Inverse of  $R$  is symmetric**

29. For  $(2x-3, 4y+2) = (1,10)$ . What will be the value of  $x$  and  $y$ ?

**(2,2)**

30. Let  $f$  and  $g$  be the two functions from  $\mathbb{R}$  to  $\mathbb{R}$  defined by  $f(x) = |x|$  and  $g(x) = \text{square root of } x^2$  for all  $x \in \mathbb{R}$ . Then \_\_\_\_\_

**$f(x)$  is not equal to  $g(x)$**

31. If a set  $A$  has 15 elements then  $P(A)$  (power set of  $A$ ) has \_\_\_\_\_ elements.

**$2^{15}$**

32. For the relation below to be a function, x cannot be what values  $\{(12,14),(13,5),(-2,7),(x,13)\}$ ?

**X cannot be 12, 13, or -2**

33. Let the set  $A = \{1,2,3,4\}$ . Then the relation  $\{(2,4),(4,2)\}$  is \_\_\_\_\_

**Symmetric**

34. For the following relation to be a function, x can not be what values?  $R=\{(2,4),(x,1), (4,2),(5,6)\}$ .

**x cannot be 2,4 and 5**

35. Vertical line test is used to determine that whether the graph of a relation is a function or not.

**True**

36. The properties of being symmetric and being anti symmetric are \_\_\_\_\_

**Not negative of each other**

37. The number of elements in  $A \times B$  are \_\_\_\_\_ if A is a set with '5' elements and B is a set with '4' elements.

**20**

38.  $R=\{(a,1)(b,2)(c,3)(d,4)\}$  then the inverse of this relation is \_\_\_\_\_

**$\{(1,a)(2,b)(3,c)(4,d)\}$**

39. Logic gate NOT does not define a binary operation on  $(0,1)$  because \_\_\_\_\_

**It takes a single input and gives a single output**

40. How many real numbers exist between 1 and 5

**3**

41. The number pi is

**Irrational**

42. The number square root 2 is

**Irrational**

43. Range the relation  $\{(0,1)(3,22),(90,34)\}$

**$\{0,3,90\}$**

44. Supherical coordinate 0 is related to the cylindrical coordinate as \_\_\_\_\_

**$@=@$**

45. Operation of subtraction is a binary operation on the set \_\_\_\_\_

**Integers**

46. Let  $A = \{1,2,3,4\}$  and  $R=\{(1,2),(2,3),(3,3),(3,4)\}$  be a relation on A. Then which one of the following ordered pair has made R not an irreflexive relation?

**$(3,3)$**

47. Input values of the function are called the \_\_\_\_\_

**Domain**

48. Range of function  $f(x)=|x|$  will be

**Set of positive real numbers**

49. Which of the following is not a binary operation on the set of integers?

**Division**

50. In the matrix representation of an irreflexive relation all the entries in the main diagonal are \_\_\_\_\_

**0**

51. If the partition set of A is  $\{A_1, A_2\}$  then

**$A_1 \cap A_2 = \text{not empty set}$**

52. Let  $A = \{a, b\}$  then  $P(A) =$

**$\{\text{Non empty set}, \{a\}, \{b\}, \{a, b\}\}$**

53. Which relations below are not functions?

$\{(13,14),(13,5),(16,7),(18,13)\}$

54. In the directed graph of an antisymmetric relation there is \_\_\_\_\_ pair of arrows between two distinct elements of the set.

No

55. If a relation  $R = (1,1),(2,1),(2,2)$  is given then which of the following is not true about this relation

R is irreflexive

56. Let R and S be transitive relations on a set A then \_\_\_\_\_

Neither R union S is transitive nor R intersection S is transitive

57. Let  $R = \{(1,2)(3,4)(5,6)(7,8)\}$ . Domain of the inverse of the relation is \_\_\_\_\_

$\{1,3,5,7\}$

58. Let  $A = \{1,2,3,4,5\}$  and  $B = \{4,9,16,17,25\}$ . Then the relation  $R = \{(2,4),(3,9),(4,16),(5,17)\}$  The inverse of R is'

$\{(4,2),(9,3),(16,4),(17,3)\}$

59. Let R be a relation on a set A. If R is symmetric then its compliment is \_\_\_\_\_

symmetric

60. Which is not a binary operation on the set of natural numbers N?

Subtraction

61. If a relation  $R = \{(1,1)(2,1)(1,2)(2,2)\}$  is given then which of the following is not true about this relation.

R is symmetric

62.  $R = \{(a,1)(b,2)(c,3)(d,4)\}$  then the inverse of this relation is \_\_\_\_\_

$\{(1,a)(2,b)(3,c)(4,d)\}$

63. For any set A, the Cartesian product of A and A is known as \_\_\_\_\_

Universal relation

64. Let  $A = \{p,q,r,s\}$  and define a relation R on A by  $R = \{(p,p),(p,r),(q,r),(q,s),(r,s)\}$  Then which one of the following is the correct statement about R:

R is not reflexive

65.  $A = \{1,2\}$   $B = \{3,4\}$ ,  $R = \{(1,3)(2,4)\}$ . Then the complement of R is \_\_\_\_\_.

$\{(1,4)(2,3)\}$

66. Domain of a relation symbolically written as \_\_\_\_\_.

$\text{Dom}(R) = \{a \in A \mid (a,b) \in R\}$

67. Let  $X = \{2,4,5\}$  and  $Y = \{1,2,4\}$  and R be a relation from X to Y defined by  $R = \{(2,4)(4,1)(a,2)\}$ . For what value of 'a' the relation R is a function?

5

68. Let  $A = \{1,2,3,4,5,6,7,8,9\}$ , then which of the following sets represent the partition of the set A?

$A = \{1,3,5,7,9\}$ ,  $B = \{2,4,6\}$ ,  $C = \{8\}$

69. Let  $A = \{1,2,3\}$  and  $B = \{2,4\}$  then number of binary relations from A to B are \_\_\_\_\_.

64

70. A relation R is said to be \_\_\_\_\_ iff it is reflexive, antisymmetric and transitive.

Partial order Relation



71. Let  $f$  be a function from  $X=\{2,4,5\}$  to  $Y=\{1,2,4,6\}$  defined as:  $f=\{(2,6),(4,2),(5,1)\}$ . The range of  $f$  is \_\_\_\_\_

**$\{1,2,6\}$**

72. Let  $A=\{0,1,2\}$  and  $R=\{(0,2),(1,1),(2,0)\}$  be a relation on  $A$ . Then which of the following statement about  $R$  is true?

**$R$  is symmetric**

73. Let  $A=\{2,3,4\}$  and  $B=\{2,6,8\}$  and let  $R$  be the “divides” relation from  $A$  to  $B$  i.e for all  $(a,b)$  belong to (Cartesian product of  $A$  and  $B$ ),  $aRb$  iff  $a \mid b$  ( $a$  divides  $b$ ). Then

**$R=\{(2,2),(2,6),(2,8),(3,6),(4,8)\}$**

74. Let  $A=\{1,2,3,\dots,50\}$  and  $B=\{2,4,6,8,10\}$ . Then the Cartesian product of  $A$  and  $B$  has \_\_\_\_\_ elements.

**250**

75. In the matrix representation of an reflexive relation all the entries in the main diagonal are \_\_\_\_\_

**1**

76. Which of the following is not a type of a relation?

**Permutation**

77. Let  $X=\{2,4,5\}$  and  $Y=\{1,2,4\}$  and  $R$  be a relation from  $X$  to  $Y$  defined by  $R=\{(2,4),(4,1),(a,2)\}$ . For what value of ‘ $a$ ’ the relation  $R$  is a function ?

**5**

78. Which of the following is not a representation of a relation?

**Venn diagram**

79. Let  $A=\{1,2,3,4\}$  and define the following relations on  $A$ . Then  $R=\{(1,3),(1,4),(2,3),(2,4),(3,1),(3,4)\}$  is \_\_\_\_\_

**$R$  is irreflexive**

80. The range of  $f:X \rightarrow Y$  is also called the image of \_\_\_\_\_

**True**

81. Complementary Relation symbolically written as \_\_\_\_\_

**$R=A \times B - R=\{(a,b) \in A \times B \mid (a,b) \text{ not belong to } R\}$**

82. Let  $A=\{1,2,3,4\}$  and  $R=\{(1,1),(2,2),(3,3),(4,4)\}$  then  $R$  is \_\_\_\_\_

**All options**

83. If a relation  $R=\{(1,1),(2,1),(1,2),(2,2)\}$  is given then which of the following is not true about this relation.

**$R$  is irreflexive**

84. Which of the following logical connective is not a binary operation ?

**Implication**

85. A set may be dividend up into its disjoint subsets, such division is called \_\_\_\_\_

**Partition**

86. If  $A=\{1,2,3\}$  &  $B=\{4,5,6\}$  and  $R=\{(1,4),(2,5),(3,6)\}$  The complementary relation is \_\_\_\_\_

**$A \times B$  (difference or  $-$ )  $R$**

87. In matrix representation of a \_\_\_\_\_ relation, the diagonal entries are always 1.

**Reflexive**

88.  $R$  is not symmetric iff there are elements  $a$  and  $b$  in  $A$  such that \_\_\_\_\_

**$(a,b)$  belongs to  $R$  but  $(b,a)$  does not belong to  $R$**

89. Which relations below are functions?

$R_1 = \{(3,4), (4,5), (6,7), (8,9)\}$

$R_2 = \{(3,4), (4,5), (6,7), (3,9)\}$

$R_3 = \{(-3,4), (4,-5), (0,0), (8,9)\}$

$R_4 = \{(8,11), (34,5), (6,17), (8,19)\}$

**R1 and R3 are functions**

90. The logic gate OR and AND are unary operation on  $\{0,1\}$

**False**

91. There is atleast one loop in the graph of an irreflexive relation

**False**

92. There is atleast one loop in the graph of an reflexive relation

**True**

93. A contains 3 elements and B contains 2 elements, then number of subsets of  $A \times B$  are \_\_\_\_\_

**64**

94. Let  $A = \{1,2,3\}$  and  $B = \{0,1,2\}$  and  $C = \{a,b\}$   $R = \{(1,0), (1,2), (3,1), (3,2)\}$   $S = \{(0,b), (1,a), (2,b)\}$  composite of R and S = \_\_\_\_\_

**$\{(1,b), (1,a), (3,a), (3,b)\}$**

95. If R is transitive then the inverse relation will be transitive

**True**

96. The number of elements in the power set of P(not empty set) denoted by  $P(P(\text{not empty set}))$  is

**2**

97. The function defined from  $\mathbb{Z}$  to  $\mathbb{Z}$  as  $f(x) = 1/(x+2)(x-2)$  is not well defined because \_\_\_\_\_

**Function is not defined at  $x=2$  and  $x=-2$**

98. Rang of relation  $\{(0,1), (3,22), (90,34)\}$  is \_\_\_\_\_

**$\{1, 22, 34\}$**

99. The number of elements in the power set of P(not empty set) denoted by  $P(P(\text{not empty set}))$  is

**1**

100. Let  $R = \{(1,2), (3,4), (5,6), (7,8)\}$ . Domain of inverse of the relation is \_\_\_\_\_

**$\{2,4,6,8\}$**

101. The relation 'divides' on the set of integers is \_\_\_\_\_

**A symmetric relation**

102. Operation of subtraction is a binary operation on the set of \_\_\_\_\_

**Integers**

103. If R is transitive then the inverse relation will be transitive.

**True**

104. Let  $A = \{1,2,3,4\}$  and define the relation R on A by  $R = \{(1,2), (2,3), (3,3), (3,4)\}$ . Then \_\_\_\_\_

**R is both reflexive and irreflexive**

105. A set may be divided up into its disjoints subsets, such division is called

**Partition**

106. If a set A contains  $n$  elements then the number of elements in its power set  $P(A)$  is \_\_\_\_\_

**$2^n$**

107. Range of a relation symbolically written as \_\_\_\_\_.

**$\text{Ran } R = \{b \in B \mid (a, b) \in R\}$**

108. Let R be a binary relation on a set A. IF R is anti symmetric then \_\_\_\_\_.

**Inverse of R is anti symmetric**

109. Let A be a set with m elements and B be a set with n elements then the number of elements in  $A \times B$  are

m.n

111. Let  $A = \{1, 2, 3, 4\}$  and define the following relation on A. Then  $R = \{(1, 3), (2, 2), (2, 4), (3, 1), (4, 2)\}$  Is

R is transitive or reflexive....both

110. If  $A = \{1, 2, 3\}$  &  $B = \{4, 5, 6\}$  and  $R = \{1, 4\}, \{2, 5\}, \{3, 6\}, \{3, 4\}$ . The complementary relation is

$A \times B$  (difference or -) R

111. Let R be a relation on a set A. If R is reflexive then its complement is

irreflexive

112.  $25 \equiv 1 \pmod{3}$  means that 3 divides

25-1

113. Let R be the universal relation on a set A then which one of the following statement about R is true ?

R is reflexive, symmetric and transitive

114. Domain of a relation symbolically written as

$\text{Dom}(R) = \{a \in R \mid (a, b) \in R\}$

115. Let R be a relation on a set A. R is transitive if and only if for all  $a, b, c \in A$  then

$(a, b) \in R$  and  $(b, c) \in R$  then  $(a, c) \in R$

116. Let A be a non-empty set and  $P(A)$  the power set of A Define the 'subset' relation,  $\subseteq$  as follows for all  $X, Y \in P(A)$ ,  $X \subseteq Y \iff$  for all  $x$ , if  $x \in X$  then  $x \in Y$ . Then  $\subseteq$  is

$\subseteq$  is partial order relation

117. Define a relation  $R = \{(1, 1), (2, 2), (3, 3), (1, 3)\}$  the relation is

R is reflexive and transitive

118. Let R be a relation on a set A. R is transitive if and only if for all  $a, b, c \in A$  then.

$(a, b) \in R$  and  $(b, c) \in R$  then  $(a, c) \in R$

119. Let R and S be reflexive relations on a set A then  $R \cap S$  is reflexive

True

## Quiz 2 spring 2021 Updated

1. A function whose range consists of only one element is called

Constant function

2. The set Z of all integers is

Countable

3. One-to-one correspondence means the condition of

Both (a) and (c)

4. Let  $X = \{1, 5, 9\}$  and  $Y = \{3, 4, 7\}$ , Define a function f from X to Y such that  $f(1) = 7, f(5) = 3, f(9) =$  . Which is true f(9) to make it a one-to-one (injective) function?

4

5. What will be the fourth term of the following sequence  $1/2, 2/3, 3/4$  ?

4/5

6. The value of  $6! =$

720

7. A constant function is one to one iff its \_\_\_\_\_ is a singleton.

Domain

8. A constant function is onto iff its \_\_\_\_\_ is a singleton.

Co-domain

9. Number of one to one functions from  $X = \{a, b\}$  to  $Y = \{u, v\}$  are equal to \_\_\_\_\_.

2

10. If  $f$  is defined recursively by  $f(0) = -1$  and  $f(n+1) = f(n) + 3$ , then  $f(2) =$ \_\_\_\_\_.

5

11. A function whose inverse function exists is called a/an \_\_\_\_\_.

Invertible function

12. Let  $f(2) = 3$ ,  $g(2) = 3$ ,  $f(4) = 1$  and  $g(4) = 2$  then the value of  $f \circ g(4)$  is.....

3

13. Let  $A = \{1, 2, 3, 4\}$  and  $B = \{7\}$  then the constant function from  $A$  to  $B$  is \_\_\_\_\_.

Onto

14. Composition of a function is a commutative operation.

False

15. Composition of a function is not a commutative operation

True

16. The sum of first five whole number is \_\_\_\_\_.

10

17. If  $f$  and  $g$  are two one-to-one functions, then their composition that is  $g \circ f$  is one-to-one.

True

18. Inverse of a surjective function is always a function.

False

19. Inverse of a surjective function may not be a function

True

20. Let  $X = \{1, 2, 3, 4\}$  and  $Y = \{7, 8, 9\}$  and let  $f$  be function defined from  $X$  to  $Y$  such that  $f$  is onto then which of the following statement about  $f$  is true?

Co-domain of  $f$  must contain 3 elements

21. If  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$  are both onto functions. Then  $g \circ f: X \rightarrow Z$  is \_\_\_\_\_

onto

22. If  $f$  and  $g$  are two one-to-one functions, then their composition  $g \circ f$  is \_\_\_\_\_

One-to-one

23. If  $f: W \rightarrow X$ ,  $g: X \rightarrow Y$ , and  $h: Y \rightarrow Z$  are functions, then \_\_\_\_\_

$(h \circ g) \circ f = h \circ (g \circ f)$

24. Cardinality of positive prime numbers less than 20 is \_\_\_\_\_.

8

25. If  $f(x) = \sin^{-1}(x)$  and  $g(x) = \sin x$  then  $g \circ f(x)$  is \_\_\_\_\_.

X

26.  $0! =$ \_\_\_\_\_.

1

27. An important data type in computer programming consists of \_\_\_\_\_.

Finite sequences

28. Let  $f(x) = 2x$  and  $g(x) = x + 2$  define functions  $f$  and  $g$  from  $R$  to  $R$ , then  $(f \circ g)(x)$  is \_\_\_\_\_.

x-2

29. The total number of terms in an arithmetic series  $0+5+10+15+....50$  are \_\_\_\_\_.

11

30.  $9!/6!=$  \_\_\_\_\_

504

31. Let  $f(x)=3x$  and  $g(x)=3x-2$  define functions  $f$  and  $g$  from  $R$  to  $R$ , THEN  $(F+G)(X)$  is \_\_\_\_\_

$6x-2$

32. If  $f$  is a bijective function then  $(f^{-1}f(x))$  is equal to

X

33. A sequence whose terms alternate in sign is called an \_\_\_\_\_

Alternating sequence

34. Common ratio in the sequence "4, 16, 64, 256,..." is.....12.

4

35. 0.8181818181 is a infinite geometric series.

True

36. The word 'algorithm' refers to a step-by-step method for performing some action.

True

37. A predicate become \_\_\_\_\_ when its variables are given specific values

Sentence

38. The sum of two irrational numbers must be an irrational number.

False

39. The sum of two irrational numbers need not be irrational number

True

40. The division by zero is allowed in mathematics.

Fasle

41. The product of any two consecutive positive integers is divisible by 2

True

42. If 'n' is an odd integer then  $n^3+n$  is \_\_\_\_\_.

Even

43. For integers a,b,c, If divides b and a divides c, then a divides (a+b).

False

44. Quotient remainder theorem states that for any positive integer d, there exist unique integer q and r such that \_\_\_\_\_ and  $0 \leq r < d$

$N=d.q+r$

45. A rule that assigns a numerical value to each outcome in a simple space is called

Random variable

46. If A and B are two disjoint (mutually exclusive) events then  $P(AB)=$

$P(A) + P(B)$

47. How many ways are there to select five players from a 10 member tennis team to make a trip to a match to another school?

$C(10,5)$

48. The expectation of x is equal to

$\sum xf(x)$

49. If  $P(A \text{ intersection } B) = P(A) P(B)$  THEN THE events A and B are called

**Independent**

50. A walk that starts and ends at the same vertex is called.

**None optins**

51. How many integers from 1 through 1000 are neither multiple of 3 nor multiple of 5

**497**

52. What is the probability of getting a number greater than 4 when a die is thrown?

**3/5**

53. Eater formula for graphs is \_\_\_\_\_.

**F=e-v+2**

54.  $X+a, x+3a, x+5a, \dots$  is an \_\_\_\_\_

**Arithmetic sequence**

55. Composition of a function is a commutative operation.

**False**

56. Real valued function is a function that assigns \_\_\_\_\_ to each member of its domain.

**Only a real number**

57. Let  $X=\{1,2,3,4\}$  and a function 'f' defined on X  $f(1)=1, f(2)=2, f(3)=3, f(4)=4$  then \_\_\_\_\_

**F is an identity function**

58. A constant function is surjective if and only if \_\_\_\_\_

**The co-domain consists of a single element**

59. Cardinality means the total number of elements in a set.

**True**

60. If  $f(x)=2x$  and  $g(x)=x$  then  $g(f(x))$  is \_\_\_\_\_.

**$2x^2$**

61. Let  $f: R \rightarrow R$  is one to one function then  $c, f, c$  is not equal 0 is also one to one function.

**True**

62. Let  $X=\{1,2,3,4\}$  and a function 'f' defined from X to X by  $f(1)=1, f(2)=1, f(3)=1, f(4)=1$  then which of the following is true?

**F is a constant function**

63. If f and g are two one-to-one functions, then their composition that is gof is one-to-one.

**True**

64. Which of the following is not correct for a 'sequence'?

**A sequence is a relation whose domain is the set of natural numbers**

65.  $F(x)=x^2$  is not one to one function from R to  $R^+$

**True**

66. Let  $f: R \rightarrow R$  is one to one function then  $c, f, c$  is not equal to is also one to one function.

**True**

67. Let  $f(x)=x+2$  then  $f^{-1}(x)$  is \_\_\_\_\_

**$x-2$**

68. Let  $f(x)=x^2+1$  define functions f from R to R and  $c=2$  be any scalar, then  $c, f(x)$  is \_\_\_\_\_.

**$2x^2-1$**

69. One to one correspondence means the condition of \_\_\_\_\_.

**Both (a) and (c)**

70. A function  $F: R \rightarrow R$  defined by  $f(x) = \text{square root } x$  is a real valued function.

**False**

71. If  $g: R \rightarrow R$  defined by  $g(x)=e^2$  is a real valued function of a real variable.

**True**



72. A function  $F: \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = \log x$  is a real valued function

True

73.  $\frac{1}{2}$ , then 3<sup>rd</sup> term of sequence is \_\_\_\_\_.

$\frac{1}{2}$

74. The process of defining an object in terms of smaller versions of itself is called recursion.

True

75. Which of the following is not correct for a 'sequence'?

A sequence is a relation whose domain is the set of natural numbers

76. A set is called countable if, and only if, it is \_\_\_\_\_.

Finite and countable infinite....both

77. A set that is not countable is called \_\_\_\_\_.

Uncountable

78. A sequence whose terms alternate in sign is called an \_\_\_\_\_.

Alternating sequence

79. Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  is one to one function then  $c \cdot f$  is also one to one function for \_\_\_\_\_.

C is not equal 0

80. Let  $f(x) = x+3$  then  $f^{-1}(x)$  is \_\_\_\_\_.

$x-3$

81. Let  $f$  and  $g$  be two functions defined by  $f(x) = x+2$  and  $g(x) = 2x+1$ . Then the composition of  $f$  and  $g$  is \_\_\_\_\_.

$2x+3$

82. Number of one to one functions from  $X = \{a, b\}$  to  $Y = \{u, v\}$  are equal to \_\_\_\_\_

2

83. The fibonacci sequence is defined as  $F_0=1, F_1=1, F_k=F_{k-1}+F_{k-2}$  for all integers  $k \geq 2$  then which of the following is true for  $F_2$

$F_2 - F_1 = 2 - 1 = 1$

84.  $x+a, x+3a, x+5a, \dots$  is an \_\_\_\_\_.

Arithmetic sequence

85. Inverse of a function may not be a function.

True

86. In the following sequence  $a_k = k/(k+1)$ , for  $k=1$ ,  $a_1$  will be \_\_\_\_\_.

$\frac{1}{2}$

87. If  $f(x)=x$  and  $g(x)=-x$  are both one to one function then  $(f+g)(x)$  is also one to one function.

False

88. If  $f(x)=x$  and  $g(x)=-x$  are both one to one function then  $(f+g)(x)$  is not one to one function.

True

89. The function ' $f$ ' and ' $g$ ' are inverse of each other if and only if their composition gives \_\_\_\_\_.

Identity function

90. Which of the following set is the domain of a sequence?

Set of real numbers

91. Let  $C$  is defined as the set of all countries in the world then  $C$  is a \_\_\_\_\_.

Finite set

92. A constant function is surjective if and only if \_\_\_\_\_.

The co domain consists of a single element

93. The sum of the series  $a_1 + a_2 + a_3 + \dots$  can be written as \_\_\_\_\_.

$$\sum_{i=1}^{\infty} a_k$$

94. Inverse of a function may not be a function.

True

95. If  $F_k = F_k - 1 + F_k - 2$  then  $F_0 = 1, f_1 = 2$ , then  $F_2 =$  \_\_\_\_\_

3

96. Let  $X = \{1, 2, 3, 4\}$  and a function 'f' defined from X to X by  $f(1)=1, f(2)=1, f(3)=1, f(4)=1$  then which of the following is true?

F is a constant function

97. The composition of function is always

Associative

98. A set is countably infinite if, and only if, it has the same cardinality as the set of

Positive integers

99. Two functions 'f' and 'g' from 'X' to 'Y' are said to be equal if and only if \_\_\_\_\_.

$F(x)=g(x)$  for all 'x' belongs to X

100.  $Y=x^3$  is a graph of bijective function from R to R.

True

101. Domain and range are same for \_\_\_\_\_.

Identity function

102. Let  $X = \{1, 2, 3, 4\}$  and a function 'f' defined on X by  $f(1)=1, f(2)=2, f(3)=3$

F is an identity function

103. Composition of a function is a commutative operation.

True

104. Inverse of a surjective function is always a function.

False

105. A function whose inverse function exists is called a/an \_\_\_\_\_.

Invertible

106. Given a set X define a function I from X to X by  $i(x)=x$  from all x belonging to X. Then \_\_\_\_\_.

I is both injective and surjective

107. Let  $f: R \rightarrow R$  is a one to one function then c, f is also one to one function for.

C is not equal 0

108. Let  $X = \{1, 5, 9\}$  and  $Y = \{3, 4, 7\}$ . Define a function f from X to Y such that  $f(1)=7, f(5)=3, f(9)=$  \_\_\_\_\_. Which is true for f(9) to make it a one-to-one (injective) function?

4

109. Which of the following is not a predecessors of  $a_k$ ?

$A_{k+1}$

110. Two functions 'f' and 'g' from X to Y are said to be equal if and only if \_\_\_\_\_.

$F(x)=g(x)$  for all 'x' belongs to x

111. The two functions 'f' and 'g' are equal if \_\_\_\_\_.

$F(x) = 3x$  and  $g(x) = 6x^2 + 3x / 2x^2 + 1$  for all  $x \in R$

112. If first term of a geometric sequence is 2 and common ratio is  $\frac{1}{2}$ , then 3<sup>rd</sup> term, of sequence.

$\frac{1}{4}$

113.  $Y = \text{square root } x$  is an \_\_\_\_\_ function form  $R^+$  to R

One to one function

114.  $Y = x^2$  is an \_\_\_\_\_ function form R to  $R^+$



**NOT ONE TO ONE FUNTION**

115. If a function  $(g \circ f)(x) : X \rightarrow Z$  is defined as  $(g \circ f)(x) = g(f(x))$  for all  $x \in X$ , Then the function \_\_\_\_\_.

**(gof)**

116. If 0 is the first term and -2 be the common difference of an arithmetic series, then the sum of first five terms of series is \_\_\_\_\_.

**-20**

117. If  $f(x) = \sin^{-1}(x)$  and  $g(x) = \sin x$  then  $g \circ f(x)$  is \_\_\_\_\_.

**X**

118.  $F: X \rightarrow Y$  that is both one to one and onto is called a \_\_\_\_\_.

**Bijjective function**

119. What does 'y' denotes in a geometric sequence?

**Common ratio**

120. Let  $g$  be a function defined by  $g(x) = x + 1$ . Then the composition of  $(g \circ g)$ .

**X+2**

121. A graph of a function  $f$  is one to one iff every horizontal line intersects the graph in at most one point.

**True**

122. Which of the following is true for the following sequence?

**If n is even, then  $C_n = 2$  and if n is odd, then  $C_n = 0$**

123. The function 'f' and 'g' are inverse of each other if and only if their composition gives \_\_\_\_\_.

**Identity function**

124.  $N!$  is defined to be \_\_\_\_\_.

**The product of the integers from 1 to n**

125. Let  $A = \{1, 2, 3, 4\}$  and  $B = \{7\}$  then the constant function from A to B

**Both one to one and onto**

126. A set is called countable if, and only if, it is \_\_\_\_\_.

**Countably infinite and finite**

127. If  $f(x) = x$  and  $g(x) = -x$  are both one to one functions then  $(f+g)(x)$  is also one to one function.

**False**

128. If  $a$  is the 1st term and  $d$  be the common difference of an arithmetic sequence then the sequence is  $a, a+d, a+2d, a+3d, \dots$

**True**

129.  $x+a, x+3a, x+5, \dots$  is  $a/a_n$  \_\_\_\_\_.

**Arithmetic sequence**

130. Inverse of an injective function may not be a function.

**True**

131.  $y = x^3$  is a graph of bijective function from  $\mathbb{R}$  to  $\mathbb{R}$

**False**

132. Two functions 'f' and 'g' from  $x$  to  $y$  are said to be equal if and only if \_\_\_\_\_.

**$F(x) = g(x)$  for all 'x' belongs to  $x$**

133. Common ratio in sequence '36, 12, 4,  $4/3, \dots$ ' is \_\_\_\_\_.

**1/3**

134. A set is called finite if, and only if, is the \_\_\_\_\_ or there is -----.

**Empty set or one-to-one**

135. Let  $f(x) = x^2 - 1$  and  $g(x) = x + 1$  define functions  $f$  and  $g$  from  $\mathbb{R}$  to  $\mathbb{R}$ , then  $(f/g)x$

**x-1**

136. A graph of a function  $f$  is one to one iff every horizontal line intersects the graph in at most one point.

**True**

137. Let  $g$  be a function defined by  $g(x) = x+1$ . Then the composition of  $(g \circ g)$ .

**X+2**

138.  $F: X \rightarrow Y$  that is both one to one and onto is called a \_\_\_\_\_.

**Bijjective function**

139. What does 'r' denotes in a geometric sequence?

**Common ratio**

140. Let  $A = \{1, 2, 3, 4\}$  and  $B = \{7\}$  then the constant function from  $A$  to  $B$  is \_\_\_\_\_.

**Both one to one and onto**

141.  $n!$  is defined to be \_\_\_\_\_.

**The product of the integers from 1 to n**

142. A set is called countable if, and only if, it is \_\_\_\_\_.

**Both b and c**

143. Which of the following is the example of an alternating sequence?

**$C_n = n/n+1$  for  $n \geq 0$**

144.  $x+a, x+3a, x+5a, \dots$  is an \_\_\_\_\_.

**Arithmetic sequence**

145.  $0, -5, -10, -15, \dots$  is an \_\_\_\_\_.

**Arithmetic sequence**

146.  $5, 9, 13, 17, \dots$  is an \_\_\_\_\_.

**Arithmetic sequence**

147. An important data type in computer programming consists of \_\_\_\_\_.

**Finite sequence**

148. One-to-one correspondence means the condition of \_\_\_\_\_.

**One-one and onto ....both**

149. Cardinality means the total number of elements in a set.

**True**

150. Inverse of a function may not be a function.

**True**

151. The functions 'f' and 'g' are inverse of each other if and only if their composition gives \_\_\_\_\_.

**Identity function**

152. A set is called finite if, and only if, it is the \_\_\_\_\_ or there is \_\_\_\_\_.

**Empty set or one-to-one**

153. Let  $f$  and  $g$  be the two functions from  $R$  to  $R$  defined by  $f(x) = |x|$  and  $g(x) = \text{square root } x^2$  for all  $x \in R$ , then \_\_\_\_\_.

**$f(x)$  is not equal to  $g(x)$**

154. If  $f^{-1}(x) = 6-x/2$  then  $f^{-1}(2)$  is \_\_\_\_\_.

**2**

155. Let  $f(2)=3, g(2)=3, f(4)=1$  and  $g(4)=2$  then the value of  $f \circ g(4)$  is \_\_\_\_\_.

**3**

156. An important data type in computer programming consists of \_\_\_\_\_.

**Finite sequence**

157. Let  $f(x)=x+3$  then  $f^{-1}(x)$  is \_\_\_\_\_.

X-3

158. If  $f: X \rightarrow Y$  and  $g: Y \rightarrow Z$  are both onto function. Then  $g \circ f: X \rightarrow Z$  is \_\_\_\_\_.

One-to-one function

159. The functions 'f' and 'g' are inverse of each other if and only if their composition gives

Identity function

160. The two function 'f' and 'g' are equal if \_\_\_\_\_.

$F(x) = 3x$  and  $g(x) = 6x^2 + 3x/2x + 1$  for all  $x \in \mathbb{R}$

161. Two functions 'f' and 'g' from X To Y are said to be equal if and only if-----.

$F(x)$  and  $g(x)$  for all 'x' belongs to X

162. Which of the following set is the domain of a sequence?

Set of natural numbers

163. If 1<sup>st</sup> term of a geometric sequence is 2 and common ratio is  $1/2$ , then 3<sup>rd</sup> term of sequence is \_\_\_\_\_.

$1/2$

164. Composition of a function is a commutative operation.

True

165. If a function  $(g \circ f)(x): X \rightarrow Z$  is defined as  $(g \circ f)(x) = g(f(x))$  for all  $x \in X$ . Then the function \_\_\_\_\_ is known as composition of f and g.

$(g \circ f)$

166. A set is countably infinite if and only if and only if, it has the same cardinality as the set of \_\_\_\_\_

Positive integers

167. The 3<sup>rd</sup> term of the sequence  $b_n = 5^n$  is \_\_\_\_\_.

125

168. If f is a bijective function then  $(f^{-1}(f(x)))$  Is equal to \_\_\_\_\_.

X

169. Let  $f(x) = x$  and  $g(x) = -x$  for all  $x \in \mathbb{R}$ , then  $(f+g)(x)$  is \_\_\_\_\_.

0

170. An infinite sequence may have only a finite number of values.

True

171. The functions  $f \circ g$  and  $g \circ f$  are always equal .

False

172. If f and g are two one-to-one functions, then their composition that is  $g \circ f$  is one-to-one.

True

173. A function whose range consists of only one element is called \_\_\_\_\_.

Constant function

174. Let  $X = \{1, 5, 9\}$  and  $Y = \{3, 4, 7\}$ . Define a function f from X to Y such that  $f(1)=7$ ,  $f(5)=3$ ,  $f(9)=4$  then which of the following statement about 'f' is true?

F is both one-to-one and onto

175.  $Y = x^3$  is a graph of bijective function from  $\mathbb{R}$  to  $\mathbb{R}$ .

True

176. A function whose inverse function exists is called a/an \_\_\_\_\_.

Invertible

177. Let F and g be two functions defined by  $f(x) = x+2$  and  $g(x) = 2x+1$ . Then the composition of f and g is \_\_\_\_\_.

$2x + 5$

178. If r is a positive real number, then the value of r in  $3, r, -27r$  is

179. The \_\_\_\_\_ of the terms of a sequence forms a series.

Sum

180. The sum of first five whole number is \_\_\_\_\_.

10

181. If  $f_k=f_{k-1}+f_{k-2}$  then  $f_0=1, f_1=2$ , then  $f_2=$  \_\_\_\_\_.

3

182. Let  $f(x) = 3x$  and  $g(x) = x + 2$  define functions  $f$  and  $g$  from  $R$  to  $R$ , then  $(f.g)(x)$  is \_\_\_\_\_.

$3x^2 + 6x$

183. Let  $R$  be a relation on a set  $A$ . If  $R$  is reflexive then its compliment is \_\_\_\_\_.

Irreflexive

184. If  $A =$  Set of students of virtual university then  $A$  has been written in the \_\_\_\_\_.

Descriptive form

185. If a function  $(g \circ f)(x):X \rightarrow Z$  is defined as  $(g \circ f)(x) = g(f(x))$  for all  $x \in X$ . Then the function \_\_\_\_\_ is known as composition of  $f$  and  $g$ .

$(g \circ f)$

186. If  $X$  and  $Y$  are independent random variables and  $a$  and  $b$  are constants, then  $Var(aX + bY)$  is equal to \_\_\_\_\_.

$aVar(X) + bVar(Y)$

187. Let  $A = \{1, 2, 3\}$  and  $B = \{2, 4\}$  then number of functions from  $A$  to  $B$  are \_\_\_\_\_.

8

188.  $p$  is equivalent to  $q'$  means \_\_\_\_\_.

$p$  is necessary and sufficient for  $q$ .

189. Let  $A$  and  $B$  be subsets of  $U$  with  $n(A) = 12, n(B) = 15, n(A') = 17$ , and  $n(A \cap B) = 8$ , then  $n(U)=$ \_\_\_\_\_.

29

190. For the following relation to be a function,  $x$  can not be what values?  
 $R = \{(2,4), (x,1), (4,2), (5,6)\}$

$x$  cannot be 2, 4 or 5

191. Find the number of the word that can be formed of the letters of the word “ELEVEN”.

120

192. There are three bus lines between  $A$  and  $B$ , and two bus lines between  $B$  and  $C$ . Find the number of ways a person can travel round trip by bus from  $A$  to  $C$  by way of  $B$

6

193. Among 20 people, 15 either swim or jog or both. If 5 swim and 6 swim and jog, how many jog?

16

194. A predicate becomes \_\_\_\_\_ when its variables are given specific values.

statement

195. Find the number of distinct permutations that can be formed using the letters of the word "BENZENE"

420

196. Suppose there are 8 different tea flavors and 5 different biscuit brands. A guest wants to take one tea and one brand of biscuit. How many choices are there for this guest?

40

197. In how many ways a student can choose one of each of the courses when he is offered 3 mathematics courses, 4 literature courses and 2 history courses.

24

198. If  $p \leftrightarrow q$  is True, then \_\_\_\_\_.

p and q both are True.

199. If A and B be events with  $P(A) = 1/3$ ,  $P(B) = 1/4$  and  $P(A \cap B) = 1/6$ , then  $P(A \cup B) =$  \_\_\_\_\_ .

5/12

200. An integer n is a perfect square if and only if \_\_\_\_\_ for some integer k.

$n = k^2$

201. If A and B are disjoint finite sets then  $n(A \cup B) =$  \_\_\_\_\_ .

$n(A) + n(B)$

202. Let  $X = \{2, 4, 5\}$  and  $Y = \{1, 2, 4\}$  and R be a relation from X to Y defined by  $R = \{(2,4), (4,1), (a,2)\}$ . For what value of 'a' the relation R is a function ?

5

203.  $\sim(P \rightarrow q)$  is logically equivalent to \_\_\_\_\_.

$p \wedge \sim q$

204. A tree is normally constructed from \_\_\_\_\_.

left to right

205. A Random variable is also called a \_\_\_\_\_.

Chance Variable

206. The conjunction  $p \wedge q$  is True when \_\_\_\_\_.

p is True, q is True

207. The logical statement  $p \wedge q$  means \_\_\_\_\_.

**p AND q**

208. Which of the followings is the factorial form of  $5 \cdot 4$ ?

**$5!/3!$**

209. What is the minimum number of students in a class to be sure that two of them are born in the same month?

**13**

210. If p is false and q is true, then  $\sim p \leftrightarrow q$  is \_\_\_\_\_.

**True**

211. If f and g are two one-to-one functions, then their composition that is gof is one-to-one.

**TRUE**

212.  $(p \vee \sim p)$  is the \_\_\_\_\_.

**Tautology**

213.  $(-2)! =$  \_\_\_\_\_ ?

**Undefined**

214. If p = It is raining, q = She will go to college  
"It is raining and she will not go to college"  
will be denoted by

**$p \wedge \sim q$**

215. Let  $X = \{1, 2, 3\}$ , then 2-combinations of the 3 elements of the set X are \_\_\_\_\_?

**$\{1, 2\}, \{1, 3\}$  and  $\{2, 3\}$**

216. Let  $f(x) = x^2 + 1$  define functions f from R to R and  $c = 2$  be any scalar, then  $c \cdot f(x)$  is \_\_\_\_\_.

**$2x^2 + 2$**

217. The disjunction of p and q is written as \_\_\_\_\_.

**$p \vee q$**

218. If X and Y are independent random variables, then  $E(XY)$  is equal to

**$E(x)E(y)$**

219. How many possible outcomes are there when a fair coin is tossed four times?

**16**

220. Which of the followings is the product set  $A * B * C$  ?where  $A = \{a\}$ ,  $B = \{b\}$ , and  $C = \{c, d\}$ .

**$\{(a, b, c), (a, b, d)\}$**

221. The number of the words that can be formed from the letters of the word, "COMMITTEE" are

$9! / (2!2!2!)$

222. One-to-One correspondence means the condition of \_\_\_\_\_.

One-One and onto

223. The functions  $f \circ g$  and  $g \circ f$  are always equal.

FALSE

224. If order matters and repetition is allowed, then which counting method should be used in order to select 'k' elements from a total of 'n' elements?

K-Sample

225. Determine values of x and y, where  $(2x, x + y) = (8, 6)$ .

$x = 4$  and  $y = 2$

226. Let g be a function defined by  $g(x) = x + 1$ . Then the composition of  $(g \circ g)(x)$  is \_\_\_\_\_.

$x + 2$

227. What is the truth value of the sentence?

'It rains if and only if there are clouds.'

False

228. Reductio and absurdum' is another name of \_\_\_\_\_.

proof by contradiction

229. x belongs to A or x belongs to B, therefore x belongs to \_\_\_\_\_.

A union B

230. Which of the followings is the product set  $A * B * C$  ? where  $A = \{a\}$ ,  $B = \{b\}$ , and  $C = \{c, d\}$ .

$\{(a, b, c), (a, b, d)\}$

231. Real valued function is a function that assigns \_\_\_\_\_ to each member of its domain.

only a real number

232. The negation of "Today is Friday" is

Today is not Friday

233. A non-zero integer d divides an integer n if and only if there exists an integer k such that \_\_\_\_\_

$n = d k$

234. The statement  $p \rightarrow q$  is logically equivalent to  $\sim q \rightarrow \sim p$

True



235. Let R be the universal relation on a set A then which one of the following statement about R is true?

R is reflexive, symmetric and transitive.

236. Let  $f(x)=3x$  and  $g(x) = 3x - 2$  define functions f and g from R to R. Then  $(f+g)(x) =$  \_\_\_\_\_.

$6x - 2$

237. The switches in parallel act just like \_\_\_\_\_.

OR gate

238. The converse of the conditional statement  $p \rightarrow q$  is

$q \rightarrow p$

239. If X and Y are random variables, then  $E(aX)$  is equal to

$aE(X)$

240. Which of the following statements is true according to the Division Algorithm?

$17 = 5 \times 3 + 2$

241. Let  $p \rightarrow q$  be a conditional statement, then the statement  $q \rightarrow p$  is called \_\_\_\_\_.

Converse

242. The disjunction  $p \vee q$  is False when \_\_\_\_\_.

p is False, q is False.

243. A student can choose a computer project from one of the two lists. The two lists contain 12 and 18 possible projects, respectively. How many possible projects are there to choose from?

30

244. The converse of the conditional statement 'If I live in Quetta, then I live in Pakistan' is \_\_\_\_\_.

If I live in Pakistan, then I live in Quetta.

245. The functions 'f' and 'g' are inverse of each other if and only if their composition gives \_\_\_\_\_.

identity function

246.  $P(0, 0)=$ \_\_\_\_\_?

1

247. Let  $p_1, p_2, p_3$  be True premises in a given Truth Table. If the conjunctions of the Conclusion with each of  $p_1, p_2, p_3$  are True, then the argument is \_\_\_\_\_.

Valid

248. If p is false and q is false, then  $\sim p$  implies q is \_\_\_\_\_.



False

249. A box contains 5 different colored light bulbs. Which of the followings is the number of ordered samples of size 3 with replacement?

125

250. Let  $A = \{2, 3, 5, 7\}$ ,  $B = \{2, 3, 5, 7, 2\}$ ,  $C =$  Set of first five prime numbers. Then from the following which statement is true ?

$A = B$

251. The set of prime numbers is \_\_\_\_\_.

infinite set

252. The contrapositive of the conditional statement 'If it is Sunday, then I go for shopping' is \_\_\_\_\_.

I do Not go for shopping, then it is Not Sunday.

253. Let  $p$  be True and  $q$  be True, then  $(\sim p \wedge q)$  is \_\_\_\_\_.

False

254. In how many ways a student can choose a course from 2 science courses, 3 literature courses and 5 art courses.

30

255. The method of loop invariants is used to prove \_\_\_\_\_ of a loop with respect to certain pre and post-conditions.

correctness

256. A student is to answer five out of nine questions on exams. Find the number of ways that can choose the five questions

126

257. If  $A$  and  $B$  are any two sets, then  $A - B = B - A$

False

258. There are 5 girls students and 20 boys students in a class. How many students are there in total ?

25

2<sup>nd</sup> file confirm

Question No. 1:

If      p= It is raining q = She will go to college

"It is raining and she will not go to college" will be denoted by

$p \wedge \sim q$

$p \wedge q$

$\sim p \wedge q$

Question No. 2:

The negation of "Today is Friday" is

Today is Saturday

Today is not Friday

Today is Thursday

None of given

Question No. 3:

The converse of the conditional statement  $p \rightarrow q$  is

$q \rightarrow p$

$\sim q \rightarrow \sim p$

$\sim p \rightarrow \sim q$

None of these

Question No. 4:

Contra-positive of given statement "If it is raining, I will take an umbrella" is

I will not take an umbrella if it is not raining. (Pg 19)

I will take an umbrella if it is

raining.

It is not raining or I will take an umbrella.

None of these.

Question No. 5:

A statement is also referred to as a

Proposition (pg 4)

Conclusion

Order

None of these

Question No. 6:

The statement "It is not raining if and only if roads are dry" is logically equivalent to

If roads are dry then it is not raining.

None of these.

Roads are dry if and only if it is not raining

If it is not raining then roads are dry.

Question No. 7:

The statement  $\sim (\sim p) = p$  Describes

Commutative Law

Implication Laws

Double negative law

Equivalence

Question No. 8:

An arrangement of rows and columns that specifies the truth value of a compound proposition for all possible truth values of its constituent propositions is called

Truth Table

Venn diagram

False Table

None of these

Question No. 9:

An argument is \_\_\_\_\_ if the conclusion is true when all the premises are true.

Valid (Pg 25)

Invalid

False

None of

these

Question No. 10:

The row in the truth table of an argument where all premises are true is called

Valid row

Invalid row

Critical row (Pg 27)

None of

these

Question No. 11:

The statement  $p \rightarrow q \equiv (p \wedge \sim q) \rightarrow c$  describes

☐

Commutative Law

☐

Implication Laws

☐

Exportation Law

☒

Reductio ad absurdum

Question No. 12:

$p \leftrightarrow q$  is logically equivalent to  $(p \rightarrow q) \wedge (q \rightarrow p)$

☐

TRUE

☐

FALSE

Question No. 13:

According to biconditional  $1+1=3$  if and only if sky is yellow.

☒

TRUE (Pg 20)

☐

FALSE

Question No. 14:

A statement that is always true regardless of the truth values of the statement variables called Tautology.

☒

TRUE (Pg 10)

☐

FALSE

Question No. 15:

If p and q are statement variables, then the conjunction of p and q is "p and q" denoted as " $p \vee q$ ".

☐

TRUE

☒

FALSE

Question No: 3 ( Marks: 1 ) - Please choose one

For two sets A,B

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$  is called

- ▶ Distributivity of intersection over union
- ▶ Distributivity of union over intersection
- ▶ None of these
- ▶ Distributivity Law

Question No: 6 ( Marks: 1 ) - Please choose one

Check whether

$36 \equiv 1 \pmod{5}$

36 Modulus 5 = 1 remainder

$33 \equiv 3 \pmod{10}$

33 Modulus 10 = 3 remainder

▶ Both are equivalent

- ▶ Second one is equivalent but first one is not
- ▶ First one is equivalent but second one is not

Question No: 7 ( Marks: 1 ) - Please choose one

A binary relation R is called Partial order relation if

- ▶ It is Reflexive and transitive
- ▶ It is symmetric and transitive
- ▶ It is reflexive, symmetric and transitive
- ▶ It is reflexive, anti-symmetric and transitive (Pg 92)

Question No: 8 ( Marks: 1 ) - Please choose one

The order pairs which are not present in a relation, must be present in

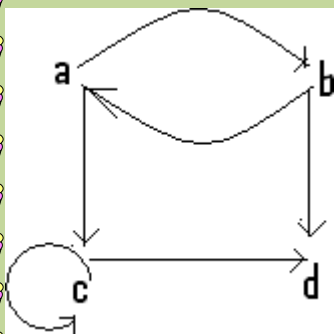
- ▶ Inverse of that relation
- ▶ Composition of relations

► Complementary relation of that relation (pg 97)

Remember own prayer

**Question No: 9 ( Marks: 1 ) - Please choose one**

The relation as a set of ordered pairs as shown in figure is



- ▶  $\{(a,b),(b,a),(b,d),(c,d)\}$
- ▶  $\{(a,b),(b,a),(a,c),(b,a),(c,c),(c,d)\}$
- ▶  $\{(a,b), (a,c), (b,a),(b,d), (c,c),(c,d)\}$
- ▶  $\{(a,b), (a,c), (b,a),(b,d),(c,d)\}$

**Question No: 10 ( Marks: 1 ) - Please choose one**

A circuit with two input signals and one output signal is called

- ▶ NOT-gate (or inverter)
- ▶ **AND- gate**
- ▶ None of these

**Question No: 11 ( Marks: 1 ) - Please choose one**

If  $f(x)=2x+1$  then its inverse =

- ▶  $x-1$
- ▶  $\frac{1}{2}(x-1)$
- ▶  $x^2 +2$

$$f(x) = 2x + 1$$

$$y = 2x + 1$$

$$x = \frac{y-1}{2}$$

$$f(x)^{-1} = \frac{y-1}{2}$$

**Question No: 12 ( Marks: 1 ) - Please choose one**

Null set is denoted by

- ▶ **(phi) or { }. (pg 39)**
- ▶ A
- ▶ None of these

**Question No: 13 ( Marks: 1 ) - Please choose one**

The total number of elements in a set is called

- ▶ Strength
- ▶ **Cardinality (pg 141)**
- ▶ Finite

Question No: 14 ( Marks: 1 ) - Please choose one

Remember own prayer

If  $f(x) = x+1$  and  $g(x) = -2x^2+1$  then  $(2f - 1g)x =$

- ▶  $2x^2 - x$
- ▶  $3x+2$
- ▶  $2x^2 + 2x+1$

$$= (2f - 1g) x$$

$$= 2f(x) - g(x)$$

$$= 2(x+1) - (-2x^2+1)$$

$$= 2x + 2 + 2x^2 - 1$$

$$= 2x^2 + 2x + 1$$

**Question No: 15 ( Marks: 1 ) - Please choose one**

Let

$$a_0 = 1, a_1 = -2 \text{ and } a_2 = 3$$

$$\text{then } \sum_{j=0}^2 a_j =$$

- ▶ -6
- ▶ 2
- ▶ 8

**Question No: 16 ( Marks: 1 ) - Please choose one**

Which of the given statement is incorrect?

- ▶ The process of defining an object in terms of smaller versions of itself is called recursion. (Pg 159)
- ▶ A recursive definition has two parts: Base and Recursion.
- ▶ Functions cannot be defined recursively (Pg 159)
- ▶ Sets can be defined recursively. (Pg 165)

**Question No: 17 ( Marks: 1 ) - Please choose one**

The operations of intersection and union on sets are commutative

- ▶ True (Pg 42)
- ▶ False
- ▶ Depends on the sets given

**Question No: 18 ( Marks: 1 ) - Please choose one**

The power set of a set A is the set of all subsets of A, denoted  $P(A)$ .

- ▶ False
- ▶ True (Pg 68)

**Question No: 19 ( Marks: 1 ) - Please choose one**

What is the output state of an OR gate if the inputs are 0 and 1?

- ▶ 0
- ▶ 1
- ▶ 2
- ▶ 3

**Question No: 20 ( Marks: 1 ) - Please choose one**

The product of the positive integers from 1 to n is called

- ▶ Multiplication
- ▶ n factorial
- ▶ Geometric sequence

**Question No: 6 ( Marks: 1 ) - Please choose one**

Let  $A = \{1, 2, 3, 4\}$  and  $R = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$  then

- ▶ R is symmetric.
- ▶ R is anti symmetric.
- ▶ R is transitive.
- ▶ R is reflexive.
- ▶ All given options are true

**Question No: 7 ( Marks: 1 ) - Please choose one**

The inverse of given relation

$R = \{(1,1),(1,2),(1,4),(3,4),(4,1)\}$  is

- ▶  $\{(1,1),(2,1),(4,1),(2,3)\}$
- ▶  $\{(1,1),(1,2),(4,1),(4,3),(1,4)\}$
- ▶  $\{(1,1),(2,1),(4,1),(4,3),(1,4)\}$  (Pg 95)

**Question No: 11 ( Marks: 1 ) - Please choose one**

$$(A \cap B)^c = (A^c \cap B^c)$$

- ▶ True
- ▶ False

**Question No: 13 ( Marks: 1 ) - Please choose one**

Let g be the functions defined by

$g(x) = 3x + 2$  then  $g \circ g(x) =$

- ▶  $9x^2 + 4$
- ▶  $6x + 4$
- ▶  $9x + 8$

$$\begin{aligned} g(g(x)) &= g(3x + 2) \\ &= 3(3x + 2) + 2 \\ &= 9x + 6 + 2 \\ &= 9x + 8 \end{aligned}$$

**Question No: 16 ( Marks: 1 ) - Please choose one**

The Common fraction for the recurring decimal 0.81 is

- ▶  $\frac{81}{100}$
- ▶  $\frac{81}{98}$
- ▶  $\frac{9}{11}$

(Pg 157)

**Question No: 17 ( Marks: 1 ) - Please choose one**

A collection of rules indicating how to form new set objects from those already known to be in the set is called

- ▶ Base
- ▶ Restriction
- ▶ Recursion (Pg159)

**Question No: 19 ( Marks: 1 ) - Please choose one**

The statement of the form  $p \vee \sim p$  is:

- ▶ Tautology (Pg 10)
- ▶ Contradiction
- ▶ Fallacy

**Question No: 20 ( Marks: 1 ) - Please choose one**

Let A,B,C be the subsets of a universal set U.

$$(A \cup B) \cup C$$

Then is equal to:

$$A \cap (B \cup C)$$



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$$A \cup (B \cap C)$$

▶  $\square$

▶  $\square \emptyset$

$$\text{▶ } A \cup (B \cup C)$$

### Associative Law

**Question:** If  $R = \{(a, a), (b, b), (c, c)\}$  is a relation on the set  $A = \{a, b, c\}$  Then  $R$  is

- Symmetric only.
- Symmetric and reflexive only.
- Reflexive only.
- **Equivalence relation. (Pg 85)**

**Question:** The negation of the implication "If  $P$  is a square then  $P$  is a rectangle" is

- **If  $P$  is not a square then  $P$  is not a rectangle**
- $P$  is not a square and  $P$  is a rectangle
- $P$  is a square and  $P$  is not a rectangle.
- None of the above

**Question:** Identify the false statement

- **$0 \in \emptyset$**
- $\{\emptyset\} \subseteq \{\emptyset\}$
- If  $A$  and  $B$  are two sets  $A \subseteq B$  and  $B \subseteq A$  then  $A = B$ .
- Two sets are disjoint if their intersection is empty set.
- $A \cap A = U$

**Question:** Let  $A$  be a set containing 3 elements then the total number of relations from  $A$  to  $A$  is

- $2^9$
- $2^{27}$
- **$n^n$**
- $2n$

**Question:** Let  $A = \{1, 2, 3\}$  and  $B = \{2, 3, 4, 5\}$  then

- $A = B$ .
- $A$  is a subset of  $B$ .
- $A$  is improper subset of  $B$ .
- **Both 2 and 3.**

**Question:** Which of the following is not a Proposition?

- **$x > 11$ .**
- Sun rounds about the Earth
- $11 + 7 = 18$
- None of above.

**Question:**  $F = \{x \in \mathbb{R} \mid x^2 + 29x + 12 = 0\}$

- **finite**
- infinite
- (c) none of above

**Question:** Let  $A$  has the same cardinality as  $B$  if and only if ,there is a ----- correspondence between sets  $A$  and  $B$

- one-one
- onto
- **(c) Both (a) and (b) (Pg 141)**

**Question:** Let  $A = \{0, 1, 2, 3, 4, 5\}$  and we define functions  $f: A \rightarrow A$  and then  $g: A \rightarrow A$

$$f(3) = 3, f(4) = 2, f(5) = 2, f(2) = 5, f(1) = 2$$

$$g(1) = 4, g(3) = 3, g(5) = 3, g(2) = 1$$

then  $f \circ g(5)$  and  $g \circ f(2)$

- **(a)  $f \circ g(5) = g \circ f(2)$**
- (b)  $f \circ g(5) \neq g \circ f(2)$
- (c)  $f \circ g(3) = g \circ f(1)$
- (d) None of the above

$$f \circ g(5) = fg(5)$$

$$= f(3) = 3$$

$$g \circ f(2) = gf(2)$$

$$= g(5) = 3$$

**Question: Choose the correct answer:**

If f and g are two one-to-one functions, then their composition of gof is

- onto
- **one-to-one (Pg 134)**
- (c ) bijective

**Question: If  $1=1$  then  $2=2$  , the conditional statement is**

- **True**
- False
- None of other.

If  $1^3 + 2^3 + 3^3 + \dots + n^3 =$

Then,

$$\left[ \frac{n(n+1)}{2} \right]^2 \quad (\text{Pg 157})$$

$$\frac{n(n+1)(2n+1)}{6}$$

$$\frac{n(n+1)}{2}$$

None of these

**Question: A set Z has n elements. How many functions are from Z to Z?**

- $2_n$
- **$n \times n$**
- $n_n$
- None of the other

**Question: Compute the summation**

$$\sum_{i=0}^2 (i^2 + 2)$$

- 5
- 3
- 0
- **None of these.**

**Question: Let  $S = \{n \in \mathbb{Z} / n = (-1)^k, \text{ for some integer } k\}$**

- $S = \{1\}$
- $S = \{-1\}$
- $S = \{-1, 1\}$
- None of the other

**Question: If  $p=T, q=T, r=F$**

Then

$$((\sim p) \wedge r) \rightarrow (q \wedge r)$$

Must be

- F
- **T**
- $q \vee r$

➤ None of these.

**Question:** If  $A=\{a,b,c,d\}$  then the number of elements of power set  $P(A)$  are

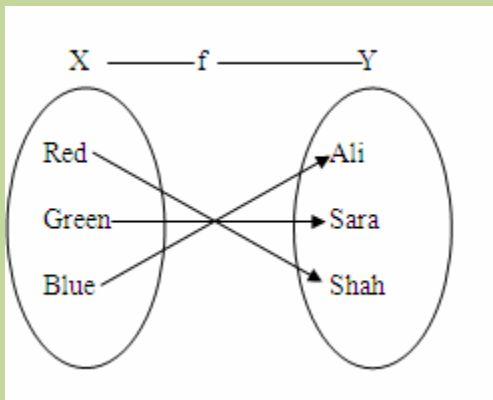
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- $2^4$
- $2^5$
- $2^6$
- $2^7$

**Question:** Consider the relation  $R=\{(1,1),(1,2),(1,4),(2,1),(2,2),(3,3),(4,4)\}$  on  $A=\{ 1,2,3,4 \}$  is

- ▶ Symmetric
- ▶ Transitive
- ▶ **Reflexive**
- ▶ All of these

**Question:** The function defined by the following diagram is  $f: X \rightarrow Y$



- ▶ One-to-one
- ▶ Onto
- ▶ **Both one-to-one and onto**
- ▶ None of these

**Question:**  $1, 10, 10^2, 10^3, 10^4, 10^5, 10^6, 10^7, \dots$  is

- ▶ **Arithmetic series (Pg 145)**
- ▶ Geometric series
- ▶ Arithmetic sequence
- ▶ Geometric sequence

**Question:**

Negations for the given statement “The train is late or my watch is fast” is

- ▶ **The train is not late or my watch is not fast.**
- ▶ The train is not late and my watch is not fast.
- ▶ The train is not late or my watch is fast
- ▶ None of these.

**Question:**

Let  $R$  be the relation from  $A=\{a_1, a_2, a_3\}$  (Elements of  $A$  are ordered by their subscript) to itself given by the matrix representation. Then  $R$  is

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

- ▶ Reflexive and Symmetric.
- ▶ Symmetric and Transitive.
- ▶ **Irreflexive and Symmetric.**
- ▶ Irreflexive and Anti- Symmetric.

If out of 35 people each person like Discrete Mathematics or Data Structures ,25 like Discrete Mathematics, and 20 like Data structures then the number of people who like

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both Discrete and Data Structures is.....

- ▶ 5
- ▶ 15
- ▶ 10
- ▶ None of these.

**Question: Inverse of a function may not be a function**

- ▶ True (Pg 124)
- ▶ False

**Question No: 1 ( Marks: 1 ) - Please choose one**

The inverse of given relation  $R = \{(1,1),(1,2),(1,4),(3,4),(4,1)\}$  is

- ▶  $\{(1,1),(2,1),(4,1),(2,3)\}$
- ▶  $\{(1,1),(1,2),(4,1),(4,3),(1,4)\}$
- ▶  $\{(1,1),(2,1),(4,1),(4,3),(1,4)\}$

**Question No: 2 ( Marks: 1 ) - Please choose one**

Symmetric and antisymmetric are

- ▶ Negative of each other
- ▶ Both are same
- ▶ Not negative of each other (Pg 90)

**Question No: 3 ( Marks: 1 ) - Please choose one**

Let  $A = \{a, b, c\}$  and

$R = \{(a, c), (b, b), (c, a)\}$  be a relation on A. Is R

- ▶ Transitive
- ▶ Reflexive
- ▶ Symmetric
- ▶ Transitive and Reflexive

**Question: In Boolean addition  $1+1=$**

- ▶ 2
- ▶ 1 (Pg 99)
- ▶ 0

**Question No: 8 ( Marks: 1 ) - Please choose one**

The same element can never appear twice in a set

- ▶ True
- ▶ False

**Question No: 9 ( Marks: 1 ) - Please choose one**

If  $f(x)=2x+1$ ,  $g(x)=x^2-1$  then  $fg(x)=$

- ▶  $x^2-1$
- ▶  $2x^2-1$

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►  $2x^3 - 1$



$$fg(x) = f(x^2 - 1)$$

$$f(x^2 - 1) = 2(x^2 - 1) + 1$$

$$= 2x^2 - 2 + 1$$

$$= 2x^2 - 1$$

Question No: 13 ( Marks: 1 ) - Please choose one

If a set contains exactly  $m$  distinct elements where  $m$  denotes some non negative integer then the set is .

- ▶ Finite
- ▶ Infinite
- ▶ None of these

Question No: 14 ( Marks: 1 ) - Please choose one

If  $f(4) = 1$  and  $g(1) = 4$  then  $f \circ g(1) =$

- ▶ 3
- ▶ 1
- ▶ 4

$$f \circ g(1) = fg(1)$$

$$= f(4) = 1$$

Question No: 15 ( Marks: 1 ) - Please choose one

If  $(A \cup B) = A$ , then  $(A \cap B) = B$

- ▶ True
- ▶ False
- ▶ Cannot be determined

Question No: 16 ( Marks: 1 ) - Please choose one

The total number of elements in a set is called

- ▶ Strength
- ▶ Cardinality
- ▶ Finite

Question No: 17 ( Marks: 1 ) - Please choose one

If  $f(x) = x$  and  $g(x) = -2x$  then  $(f+g)x =$

- ▶  $3x$
- ▶  $2x^2$
- ▶  $-x$

Question No: 19 ( Marks: 1 ) - Please choose one

Which term of the sequence  $4, 1, -2, \dots$  is  $-77$

- ▶ 26
- ▶ 27
- ▶ 28

Question No: 20 ( Marks: 1 ) - Please choose one

If a set  $A$  has 5 elements then power set of  $A$ ,  $P(S)$  contains elements. Which are?

- ▶  $5^5$
- ▶  $2^n$
- ▶  $4^5$

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2<sup>5</sup>



**Question No: 13 ( Marks: 1 ) - Please choose one**

Let g be the functions defined by

$g(x) = 3x + 2$  then  $g \circ g(x) =$

- ☐  $9x^2 + 4$
- ☐  $6x + 4$
- ☒  $9x + 8$

$$g \circ g(x)$$

$$= 3(3x + 2) + 2$$

$$= 9x + 6 + 2$$

$$= 9x + 8$$

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